

QUESTION 1

1.1) SuperNet Calculation:

Subnet Mask:

- $32 - 18 = 14$
- 18 network bits and 14 hosts
- 11111111.11111111.11000000.00000000
- 255.255.192.0

Network Address:

- 10.10.0.0/18

Broadcast Address:

- 0-63, 64-127, 128-191, 192-255
- $256 - 192 = 64$
- 10.10.0.0 -> 10.10.63.255

Total Address:

- $32 - 18 = 14$ bits
- $= 2^{14} = 16,384$

Usable hosts:

- $16,384 - 2 = 16,382$

1.2) Subnetting:

• CW Operations

$$400 \times 1.30 = 520$$

$$2^{10} - 2 = 1022$$

• CW Management:

$$100 \times 1.30 = 130$$

$$2^8 - 2 = 254$$

• DH logistics:

$$200 \times 1.30 = 260$$

$$2^9 - 2 = 510$$

- **DH security:**

$$50 \times 1.30 = 65$$

$$2^7 - 2 = 126$$

- **WAN link:**

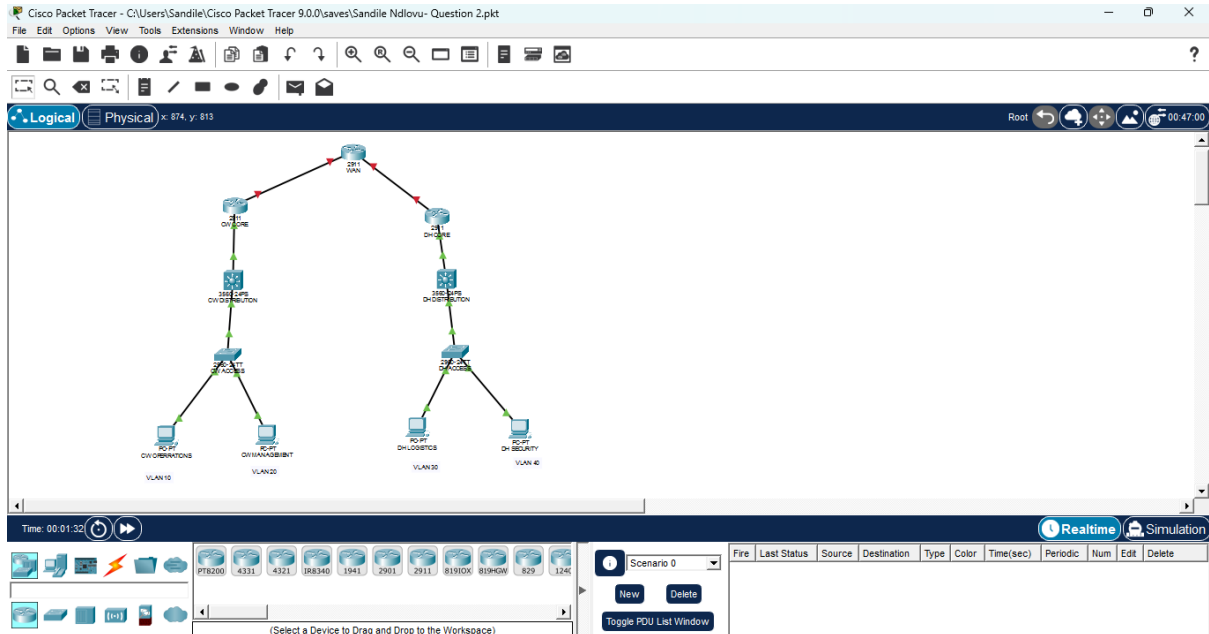
2 usable addresses

SUBNETTING TABLE

Functional Area	Host	30%Growth	Block Size	CIDR	Subnet	Usable Range	Broadcast
CW operations	400	520	1022	/22	255.255. 252.0	10.10.0.1 – 10.10.3.254	10.10.3.255
CW Management	100	130	256	/24	255.255. 255.0	10.10.6.1 – 10.10.6.254	10.10.6.255
DH logistics	200	260	512	/23	255.255. 254.0	10.10.4.1 – 10.10.5.254	10.10.5.255
DH security	50	60	128	/25	255.255. 255.128	10.10.7.1 – 10.10.7.126	10.10.7.127
WAN link	2	fixed	4	/30	255.255. 255.252	10.10.7.129 – 10.10.7.130	10.10.7.131

QUESTION 2

2.1) Topology:



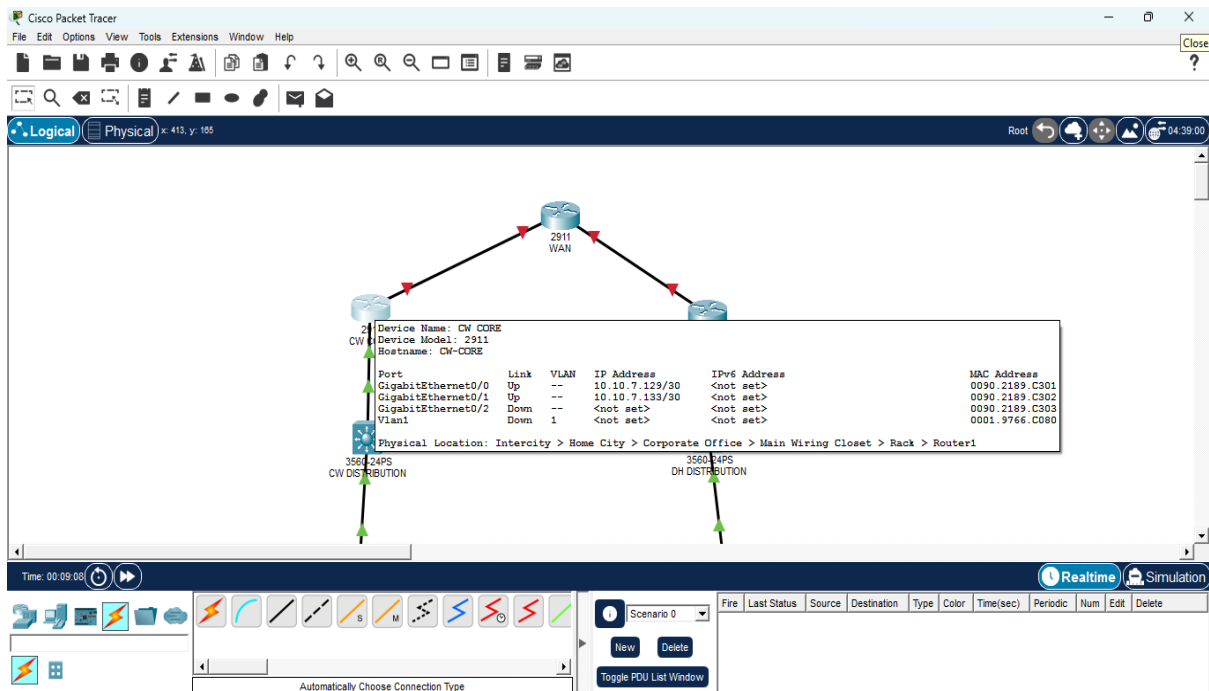
2.1a) Configuration of the CW CORE: [Router]

```
Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname CW-CORE
CW-CORE(config)#interface GigabitEthernet0/0
CW-CORE(config-if)#description WAN link to DH-CORE
CW-CORE(config-if)#ip address 10.10.7.129 255.255.255.252
CW-CORE(config-if)#no shutdown
CW-CORE(config-if)#
%LINK-6-CHANGED: Interface GigabitEthernet0/0, changed state to up
CW-CORE(config-if)#interface GigabitEthernet0/1
CW-CORE(config-if)#description link to CW-DIST
CW-CORE(config-if)#ip address 10.10.7.133 255.255.255.252
CW-CORE(config-if)#no shutdown
CW-CORE(config-if)#
%LINK-6-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
CW-CORE(config-if)#router ospf 1
CW-CORE(config-router)#router-id 1.1.1.1
CW-CORE(config-router)#network 10.10.7.128 0.0.0.3 area 0
CW-CORE(config-router)#network 10.10.7.132 0.0.0.3 area 0
CW-CORE(config-router)#end
CW-CORE#
%SYS-5-CONFIG_I: Configured from console by console

CW-CORE#write memory
Building configuration...
[OK]
CW-CORE#
```

Result of Configuration:



Configuration of the DH CORE: [Router]

```
DH CORE
Physical Config CLI Attributes
IOS Command Line Interface

Would you like to enter the initial configuration dialog? [yes/no]: n

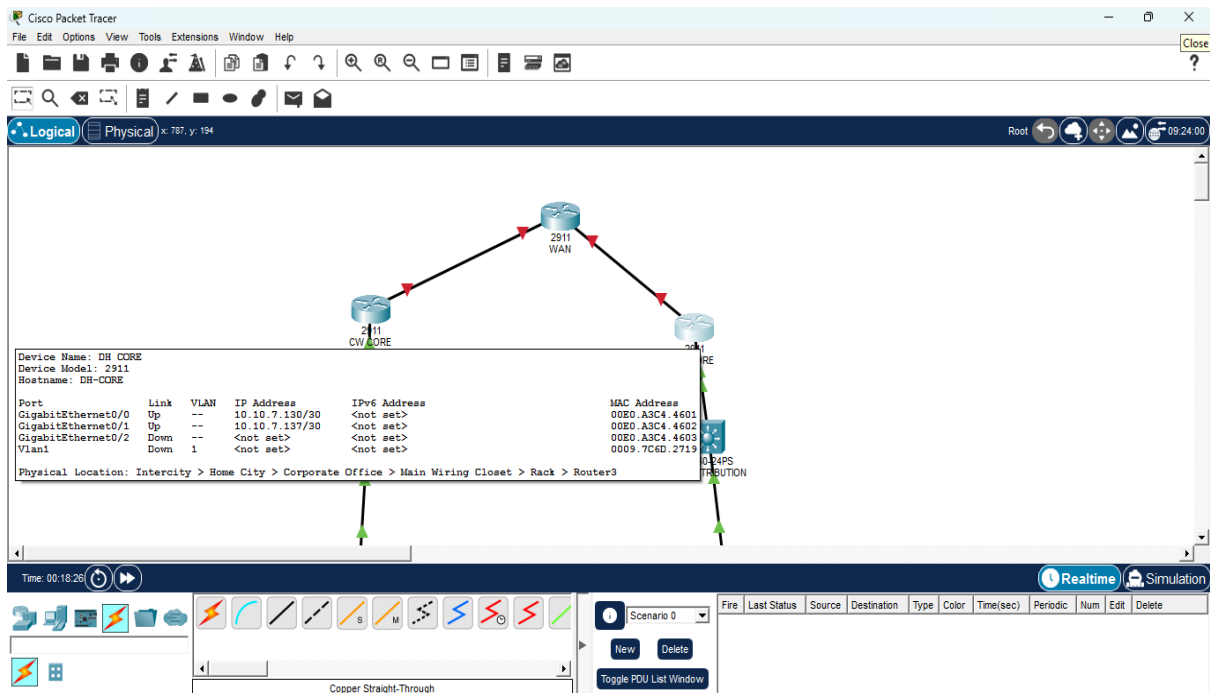
Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname DH-CORE
DH-CORE(config)#interface GigabitEthernet0/0
DH-CORE(config-if)#description WAN link to CW-CORE
DH-CORE(config-if)#ip address 10.10.7.130 255.255.255.252
DH-CORE(config-if)#no shutdown

DH-CORE(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
DH-CORE(config-if)#interface GigabitEthernet0/1
DH-CORE(config-if)#description link to DH-DIST
DH-CORE(config-if)#ip address 10.10.7.137 255.255.255.252
DH-CORE(config-if)#no shutdown

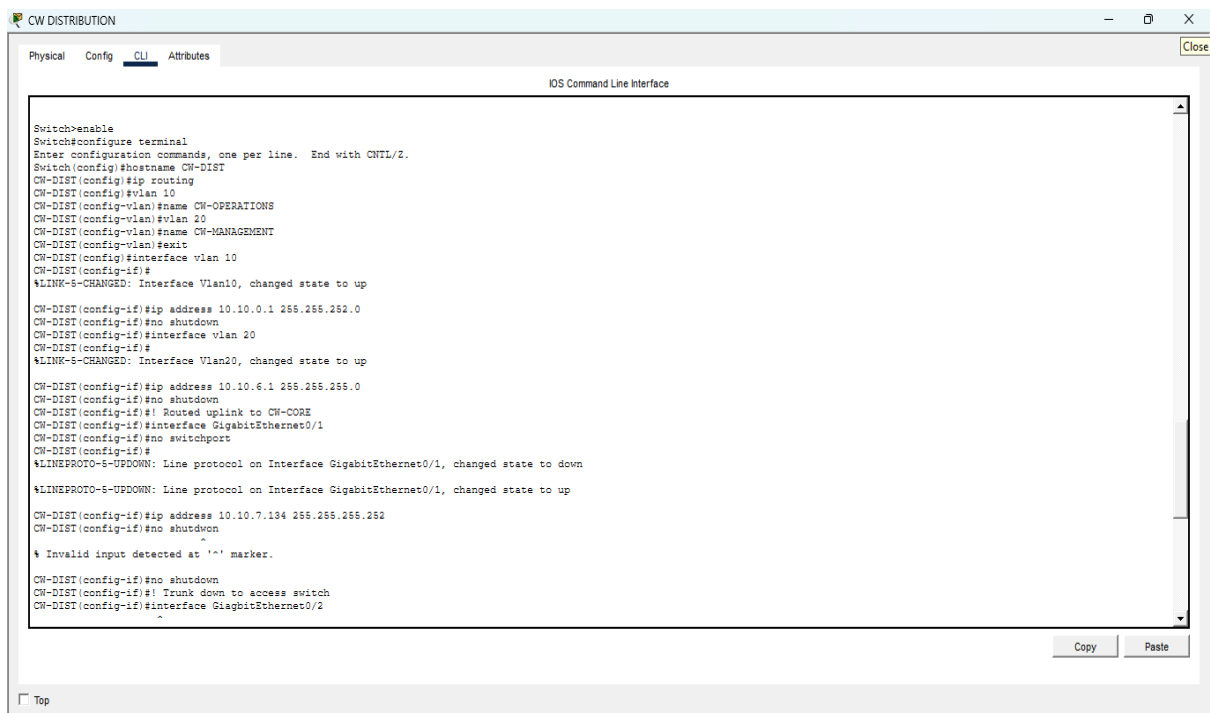
DH-CORE(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
DH-CORE(config-if)#router ospf 1
DH-CORE(config-router)#router-id 2.2.2.2
DH-CORE(config-router)#network 10.10.7.128 0.0.0.3 area 0
DH-CORE(config-router)#network 10.10.7.136 0.0.0.3 area 0
DH-CORE(config-router)#end
DH-CORE#
%SYS-5-CONFIG_I: Configured from console by console
DH-CORE#write memory
Building configuration...
[OK]
DH-CORE#
```

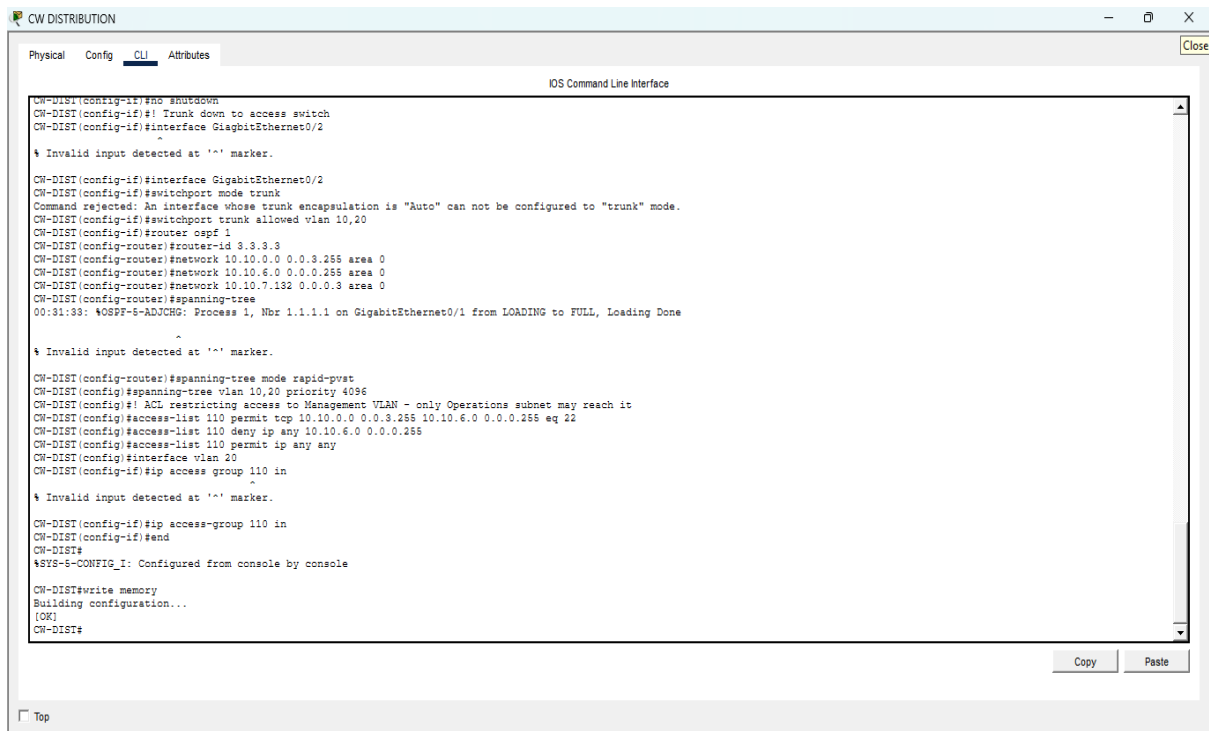
Result of Configuration:



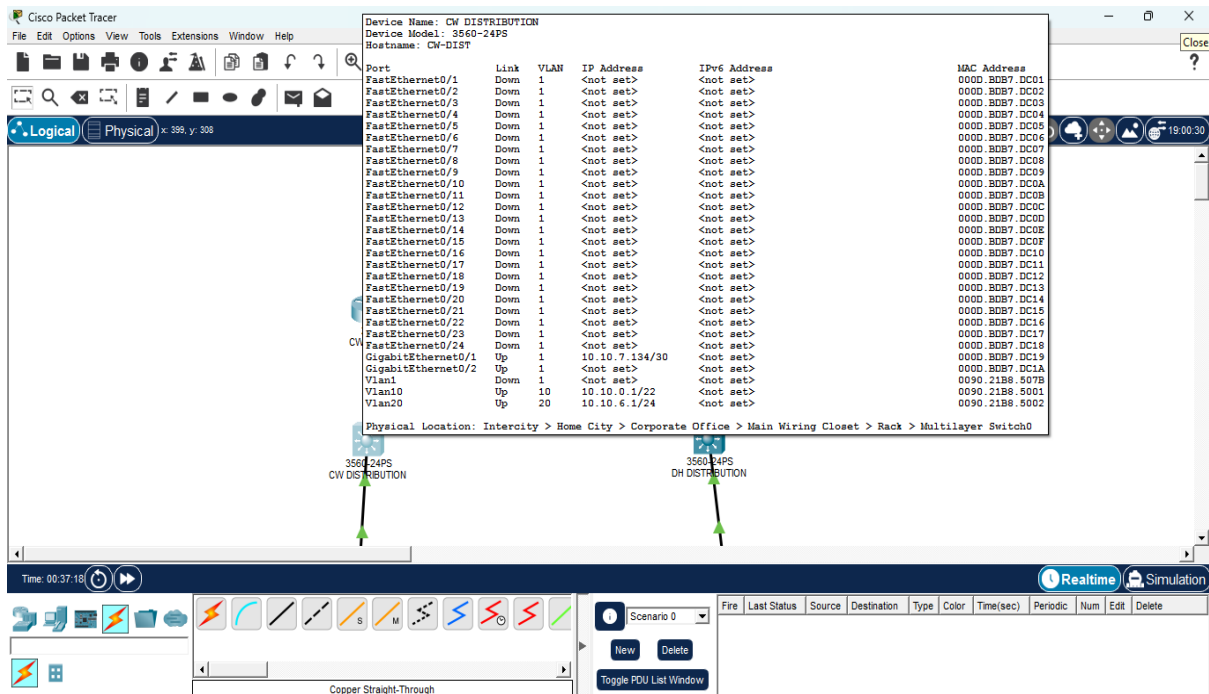
2.1b) Configuration of the Layer 3 switch

CW DISTRIBUTION:





Result of Configuration:



Configuration of the DH DISTRIBUTION LAYER:

DH DISTRIBUTION

Physical Config **CLI** Attributes

IOS Command Line Interface

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname DH-DIST
DH-DIST(config)#ip routing
DH-DIST(config)#vlan 30
DH-DIST(config-vlan)#name DH-LOGISTICS
DH-DIST(config-vlan)#vlan 40
DH-DIST(config-vlan)#name DH-SECURITY
DH-DIST(config-vlan)#exit
DH-DIST(config)#interface vlan 30
DH-DIST(config-if)#
%LINK-6-CHANGED: Interface Vlan30, changed state to up
DH-DIST(config-if)#ip address 10.10.4.1 255.255.254.0
DH-DIST(config-if)#no shutdown
DH-DIST(config-if)#interface vlan 40
DH-DIST(config-if)#
%LINK-6-CHANGED: Interface Vlan40, changed state to up
DH-DIST(config-if)#ip address 10.10.7.138 255.255.255.252
DH-DIST(config-if)#ip address 10.10.7.138 255.255.255.128
DH-DIST(config-if)#no shutdown
DH-DIST(config-if)#interface GigabitEthernet0/1
DH-DIST(config-if)#no switchport
DH-DIST(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
DH-DIST(config-if)#ip address 10.10.7.138 255.255.255/252
^
% Invalid input detected at '^' marker.
DH-DIST(config-if)#ip address 10.10.7.138 255.255.255.252
^
% 10.10.7.136 overlaps with Vlan40
DH-DIST(config-if)#no shutdown
DH-DIST(config-if)#interface GigabitEthernet0/2
```

Copy Paste

Top

DH DISTRIBUTION

Physical Config **CLI** Attributes

IOS Command Line Interface

```
DH-DIST(config-if)#ip address 10.10.7.138 255.255.255.128
DH-DIST(config-if)#no shutdown
DH-DIST(config-if)#interface GigabitEthernet0/1
DH-DIST(config-if)#no switchport
DH-DIST(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
DH-DIST(config-if)#ip address 10.10.7.138 255.255.255/252
^
% Invalid input detected at '^' marker.
DH-DIST(config-if)#ip address 10.10.7.138 255.255.255.252
^
% 10.10.7.136 overlaps with Vlan40
DH-DIST(config-if)#no shutdown
DH-DIST(config-if)#interface GigabitEthernet0/2
DH-DIST(config-if)#switchport mode trunk
Command rejected: An interface whose trunk encapsulation is "Auto" can not be configured to "trunk" mode.
DH-DIST(config-if)#switchport trunk allowed vlan 30,40
DH-DIST(config-if)#router ospf 1
DH-DIST(config-router)#router-id 4.4.4.4
DH-DIST(config-router)#network 10.10.4.0 0.0.1.255 area 0
DH-DIST(config-router)#network 10.10.7.0 0.0.0.127 area 0
DH-DIST(config-router)#network 10.10.7.136 0.0.0.9 area 0
DH-DIST(config-router)#spanning-tree mode rapid-pvst
DH-DIST(config)#spanning-tree vlan 30,40 priority 4096
DH-DIST(config)#access-list 120 permit tcp 10.10.4.0 0.0.1.255 10.10.7.0 0.0.0.127 eq 22
DH-DIST(config)#access-list 120 deny ip any 10.10.7.0 0.0.0.127
DH-DIST(config)#access-list 120 permit ip any any
DH-DIST(config)#interface vlan 40
DH-DIST(config-if)#ip access-group 120 in
DH-DIST(config-if)#end
DH-DIST#
%SYS-5-CONFIG_I: Configured from console by console
DH-DIST#write memory
Building configuration...
[OK]
DH-DIST#
```

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Top

Result of Configuration:

Device Name: DH DISTRIBUTION
Device Model: 3560-24PS
Hostname: DH-DIST

Port	Link	VLAN	IP Address	IPv6 Address	MAC Address
FastEthernet0/1	Down	1	<not set>	<not set>	00E0.8F04.9001
FastEthernet0/2	Down	1	<not set>	<not set>	00E0.8F04.9002
FastEthernet0/3	Down	1	<not set>	<not set>	00E0.8F04.9003
FastEthernet0/4	Down	1	<not set>	<not set>	00E0.8F04.9004
FastEthernet0/5	Down	1	<not set>	<not set>	00E0.8F04.9005
FastEthernet0/6	Down	1	<not set>	<not set>	00E0.8F04.9006
FastEthernet0/7	Down	1	<not set>	<not set>	00E0.8F04.9007
FastEthernet0/8	Down	1	<not set>	<not set>	00E0.8F04.9008
FastEthernet0/9	Down	1	<not set>	<not set>	00E0.8F04.9009
FastEthernet0/10	Down	1	<not set>	<not set>	00E0.8F04.900A
FastEthernet0/11	Down	1	<not set>	<not set>	00E0.8F04.900B
FastEthernet0/12	Down	1	<not set>	<not set>	00E0.8F04.900C
FastEthernet0/13	Down	1	<not set>	<not set>	00E0.8F04.900D
FastEthernet0/14	Down	1	<not set>	<not set>	00E0.8F04.900E
FastEthernet0/15	Down	1	<not set>	<not set>	00E0.8F04.900F
FastEthernet0/16	Down	1	<not set>	<not set>	00E0.8F04.9010
FastEthernet0/17	Down	1	<not set>	<not set>	00E0.8F04.9011
FastEthernet0/18	Down	1	<not set>	<not set>	00E0.8F04.9012
FastEthernet0/19	Down	1	<not set>	<not set>	00E0.8F04.9013
FastEthernet0/20	Down	1	<not set>	<not set>	00E0.8F04.9014
FastEthernet0/21	Down	1	<not set>	<not set>	00E0.8F04.9015
FastEthernet0/22	Down	1	<not set>	<not set>	00E0.8F04.9016
FastEthernet0/23	Down	1	<not set>	<not set>	00E0.8F04.9017
FastEthernet0/24	Down	1	<not set>	<not set>	00E0.8F04.9018
GigabitEthernet0/1	Up	1	<not set>	<not set>	00E0.8F04.9019
GigabitEthernet0/2	Up	1	<not set>	<not set>	00E0.8F04.901A
Vlan1	Up	1	<not set>	<not set>	0002.1691.2760
Vlan30	Up	30	10.10.4.1/23	<not set>	0002.1691.2701
Vlan40	Up	40	10.10.7.138/25	<not set>	0002.1691.2702

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > Multilayer Switch1

3560-24PS CW DISTRIBUTION

3560-24PS DH DISTRIBUTION

Time: 00:54:21

Realtime Simulation

Scenario 0

New Delete

Toggle PDU List Window

Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit Delete

Copper Straight-Through

2.1c) Configuration of the Layer 2 switch:

CW ACCESS:

CW ACCESS

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

Switch>enable
Switch>configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname CW-ACCESS
CW-ACCESS(config)#spanning-tree mode rapid-pvst
CW-ACCESS(config-vlan)#name CW-OPERATIONS
CW-ACCESS(config-vlan)#vlan 20
CW-ACCESS(config-vlan)#name CW-MANAGEMENT
CW-ACCESS(config-vlan)#exit
CW-ACCESS(config)#interface range FastEthernet0/1-15
CW-ACCESS(config-if-range)#switchport mode access
CW-ACCESS(config-if-range)#switchport access vlan 10
CW-ACCESS(config-if-range)#interface range FastEthernet0/16-20
CW-ACCESS(config-if-range)#switchport mode access
CW-ACCESS(config-if-range)#switchport access vlan 20
CW-ACCESS(config-if-range)#interface GigabitEthernet0/1
CW-ACCESS(config-if)#switchport mode trunk

CW-ACCESS(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

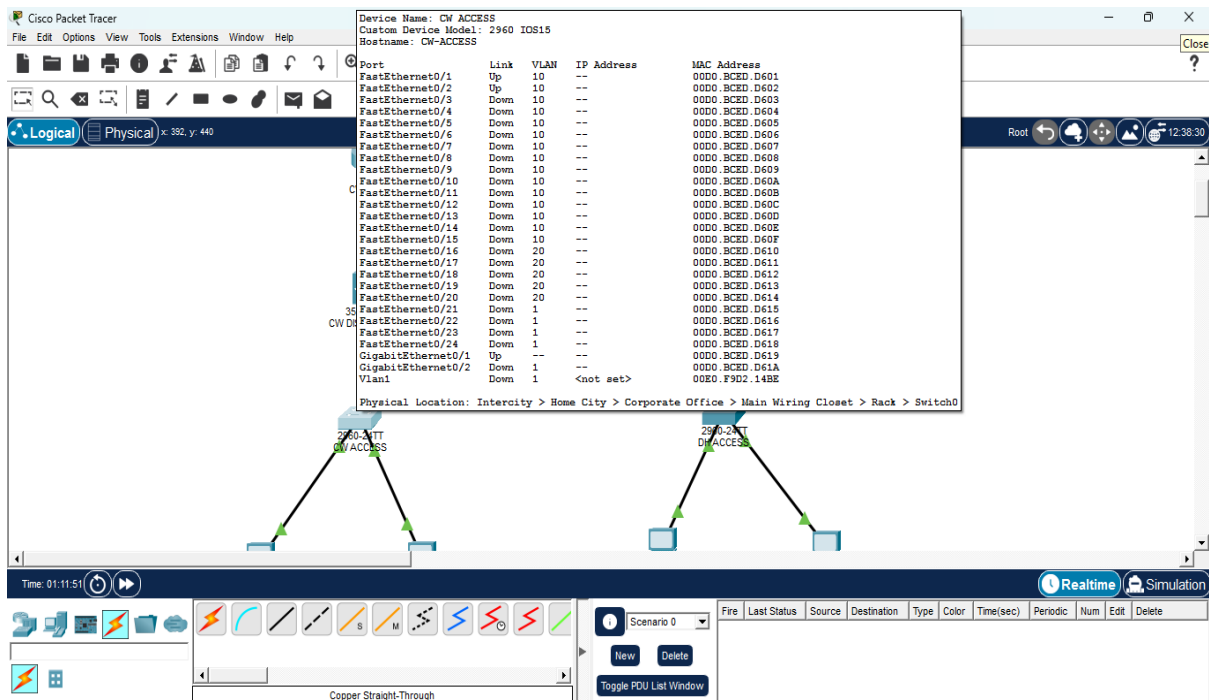
CW-ACCESS(config-if)#switchport trunk allowed vlan 10,20
CW-ACCESS(config-if)#end
CW-ACCESS#
%SYS-5-CONFIG_I: Configured from console by console

CW-ACCESS#write memory
Building configuration...
[OK]
CW-ACCESS#
```

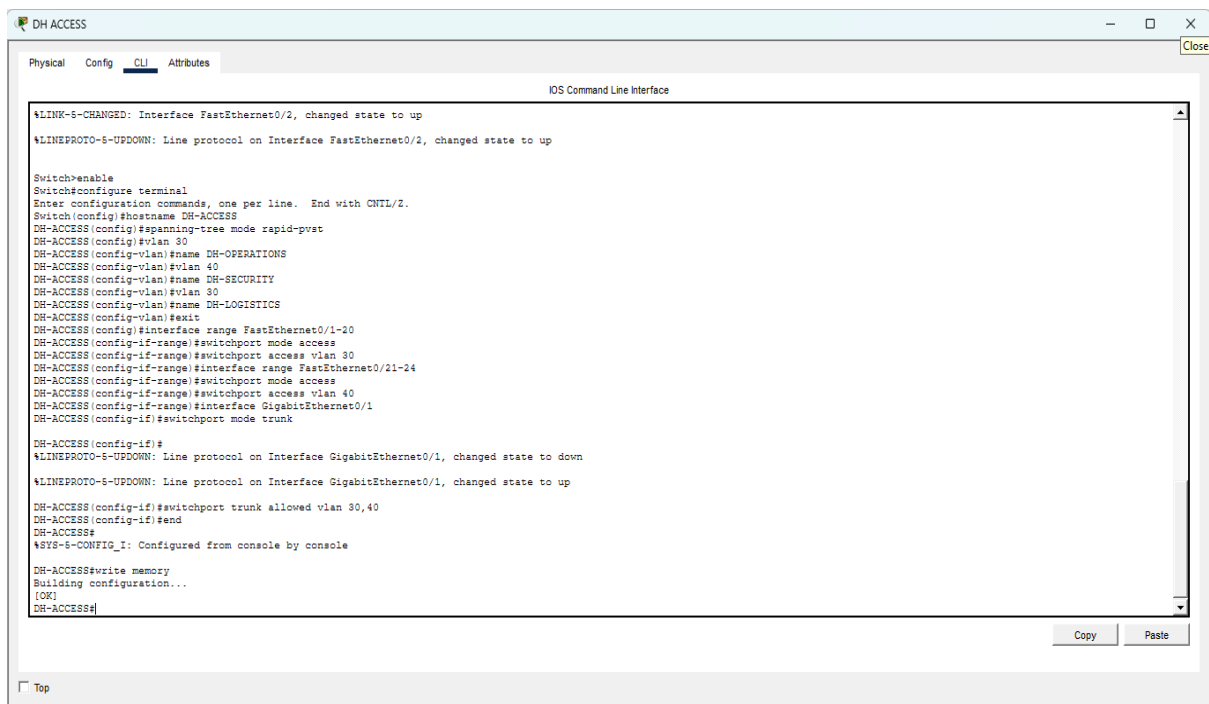
Copy Paste

Top

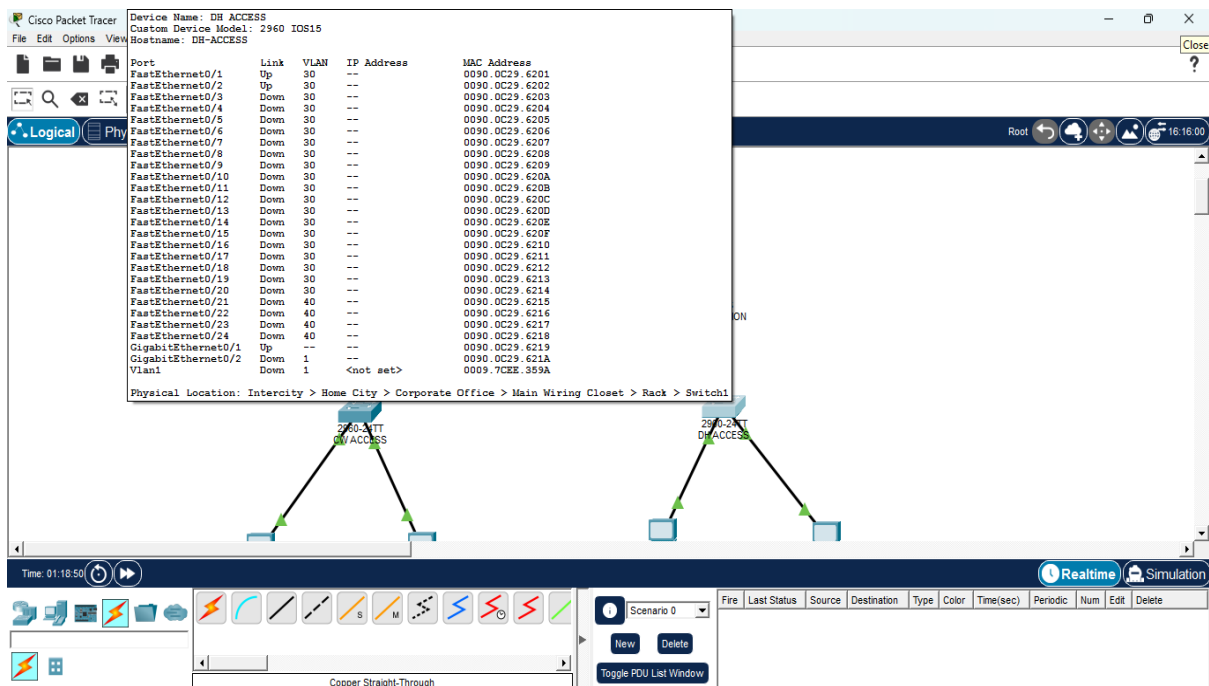
Result of Configuration:



Configuration of DH ACCESS:



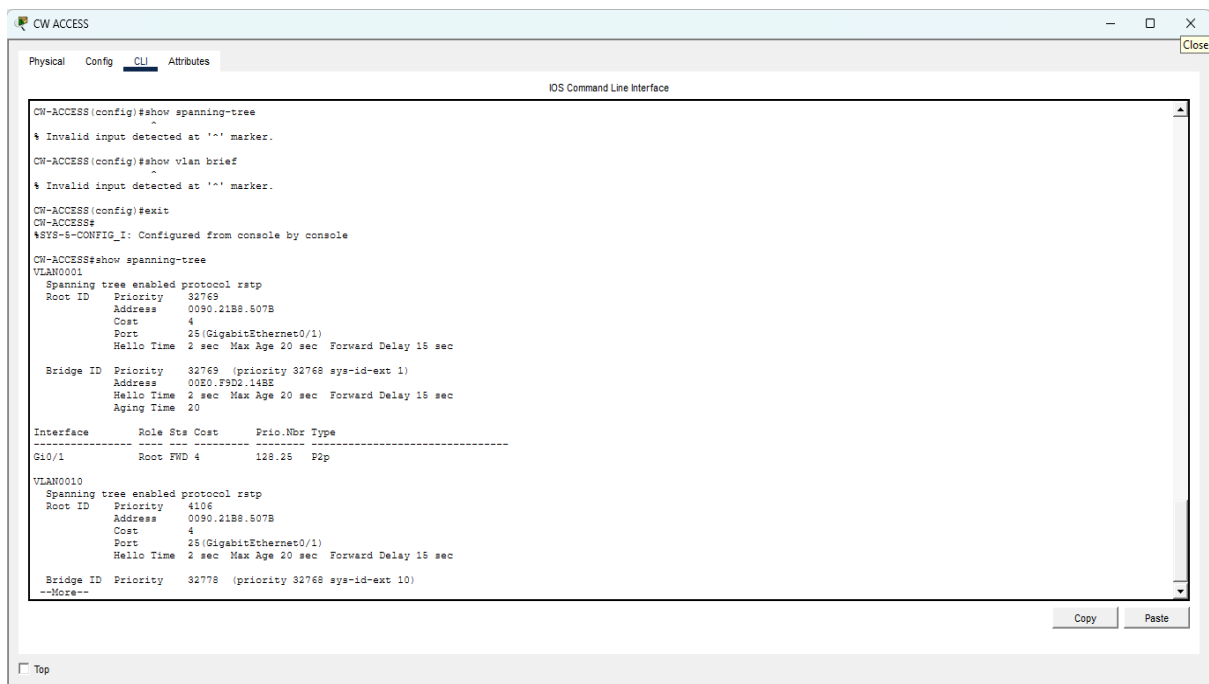
Result of Configuration



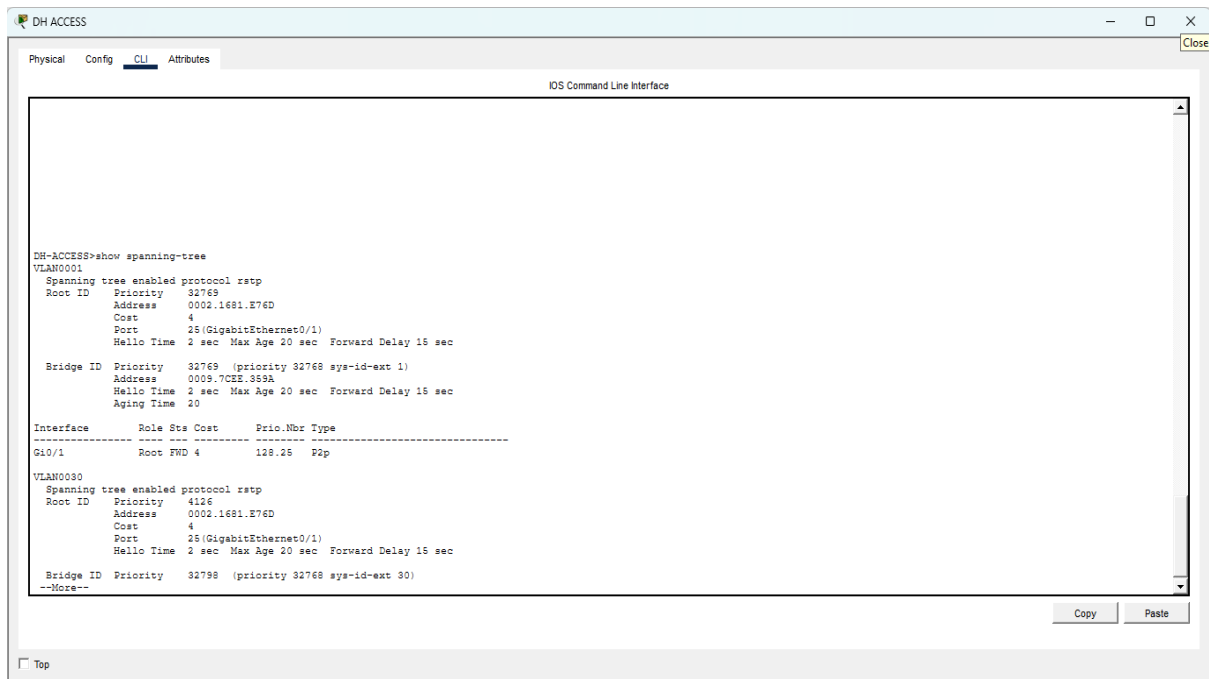
2.2b)

Configuration of the Rapid Spanning tree Protocol:

CW ACCESS:



DH ACCESS:



The screenshot shows a network configuration window titled "DH ACCESS" with tabs for Physical, Config, CLI, and Attributes. The CLI tab is active, displaying the output of the "show spanning-tree" command for two VLANs.

```
DH-ACCESS>show spanning-tree
VLAN0001
  Spanning tree enabled protocol rstp
    Root ID    Priority    32769
              Address    0002.1681.276D
              Cost        4
              Port        25(GigabitEthernet0/1)
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

    Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
              Address    0009.70CE.359A
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time  20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Gi0/1          Root FWD 4         128.25   P2p

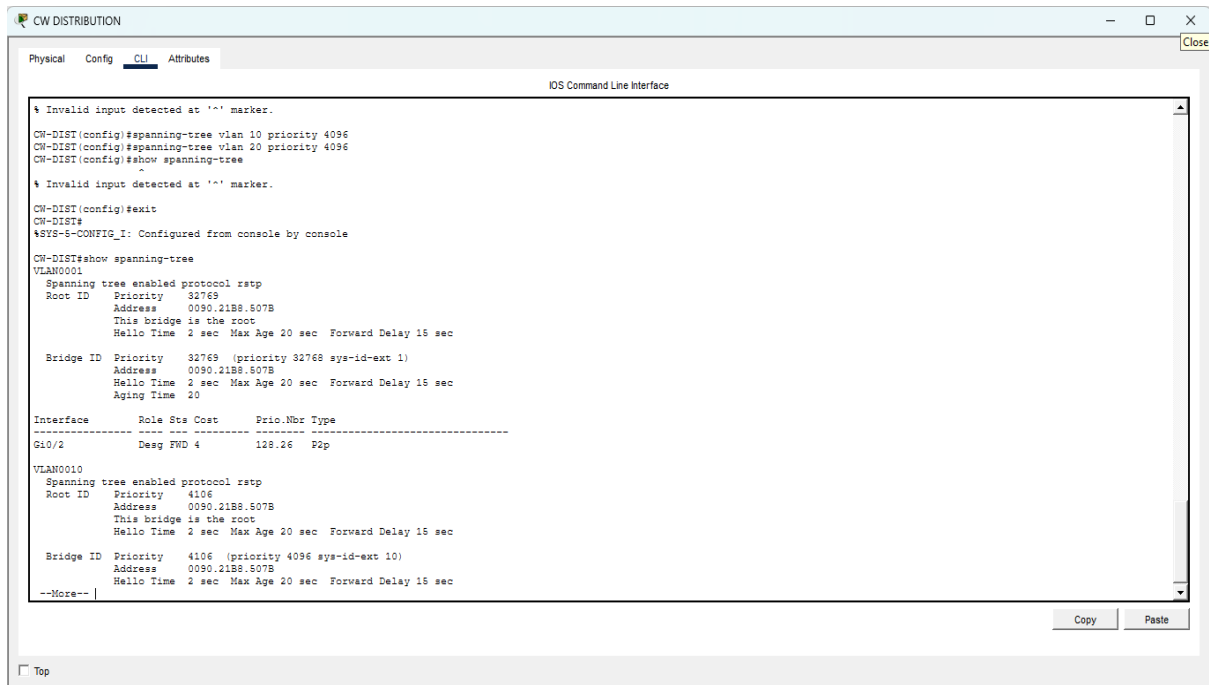
VLAN0030
  Spanning tree enabled protocol rstp
    Root ID    Priority    4126
              Address    0002.1681.276D
              Cost        4
              Port        25(GigabitEthernet0/1)
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

    Bridge ID  Priority    32798 (priority 32768 sys-id-ext 30)
              Address    0009.70CE.359A
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time  20

--More--
```

Verifying the Root Election under the Layer 3 switches:

CW DISTRIBUTION:



The screenshot shows a network configuration window titled "CW DISTRIBUTION" with tabs for Physical, Config, CLI, and Attributes. The CLI tab is active, displaying the output of the "show spanning-tree" command for two VLANs.

```
CW-DIST(config)#spanning-tree vlan 10 priority 4096
CW-DIST(config)#spanning-tree vlan 20 priority 4096
CW-DIST(config)#show spanning-tree
^
Invalid input detected at '^' marker.

CW-DIST(config)#exit
CW-DIST#
^
Invalid input detected at '^' marker.

CW-DIST#show spanning-tree
VLAN0001
  Spanning tree enabled protocol rstp
    Root ID    Priority    32769
              Address    0090.21B8.507B
              This bridge is the root
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

    Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
              Address    0090.21B8.507B
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time  20

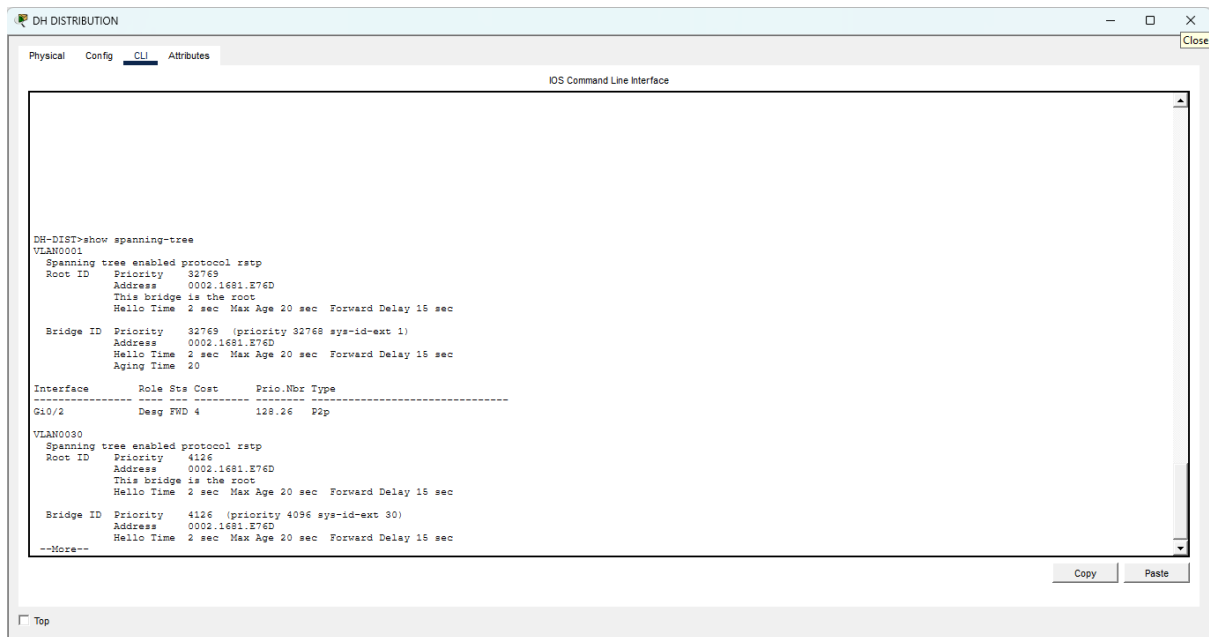
Interface      Role Sts Cost      Prio.Nbr Type
-----
Gi0/2          Desg FWD 4         128.26   P2p

VLAN0010
  Spanning tree enabled protocol rstp
    Root ID    Priority    4106
              Address    0090.21B8.507B
              This bridge is the root
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

    Bridge ID  Priority    4106 (priority 4096 sys-id-ext 10)
              Address    0090.21B8.507B
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

--More--
```

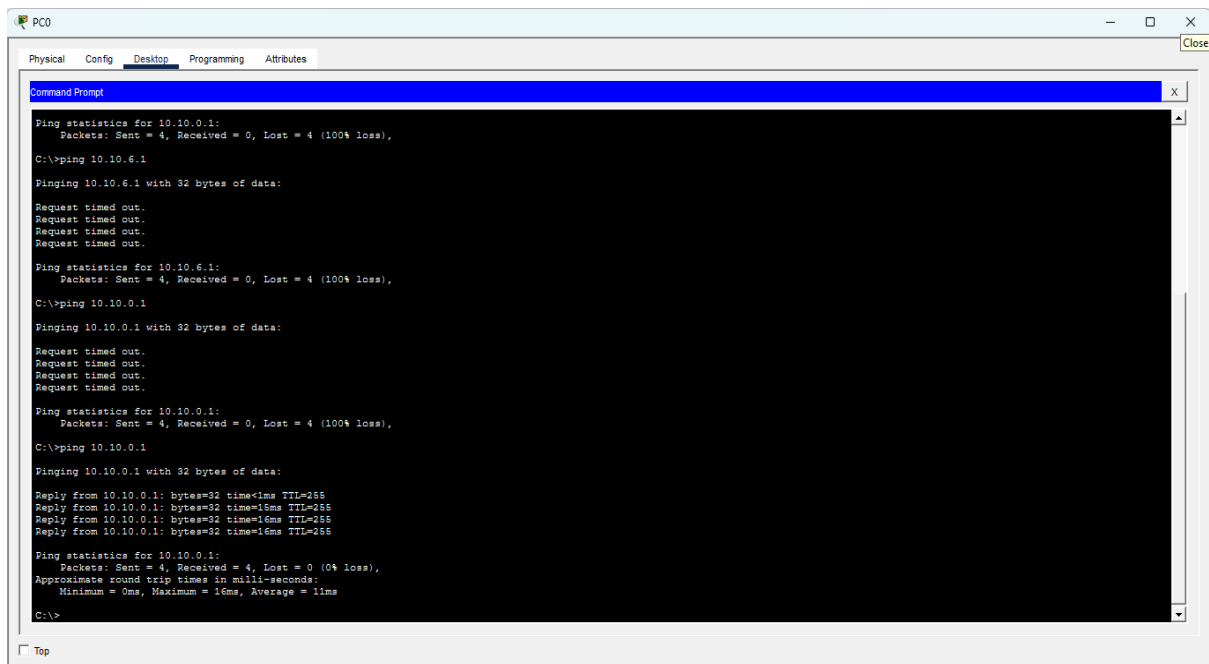
DH DISTRIBUTION:



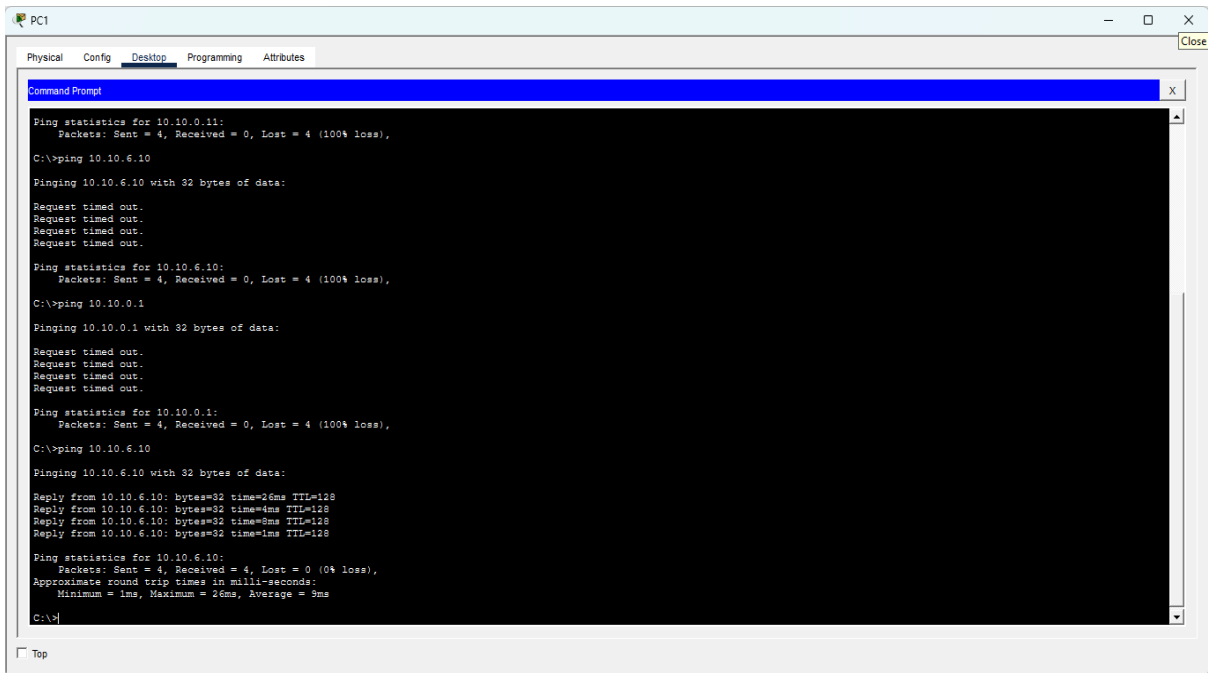
2.2c)

Traffic Flow:

The test connectivity was for VLAN 10. After a few failed attempts I was able to get the correct ping at the end



For VLAN 20 I faced the same problem. After a few failed attempts at pinging the correct IP address I finally succeeded at the end\



```
PC1
Physical Config Desktop Programming Attributes
Command Prompt
Ping statistics for 10.10.0.11:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.10.6.10

Pinging 10.10.6.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.10.6.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.10.0.1

Pinging 10.10.0.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.10.0.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.10.6.10

Pinging 10.10.6.10 with 32 bytes of data:

Reply from 10.10.6.10: bytes=32 time=26ms TTL=128
Reply from 10.10.6.10: bytes=32 time=4ms TTL=128
Reply from 10.10.6.10: bytes=32 time=8ms TTL=128
Reply from 10.10.6.10: bytes=32 time=1ms TTL=128

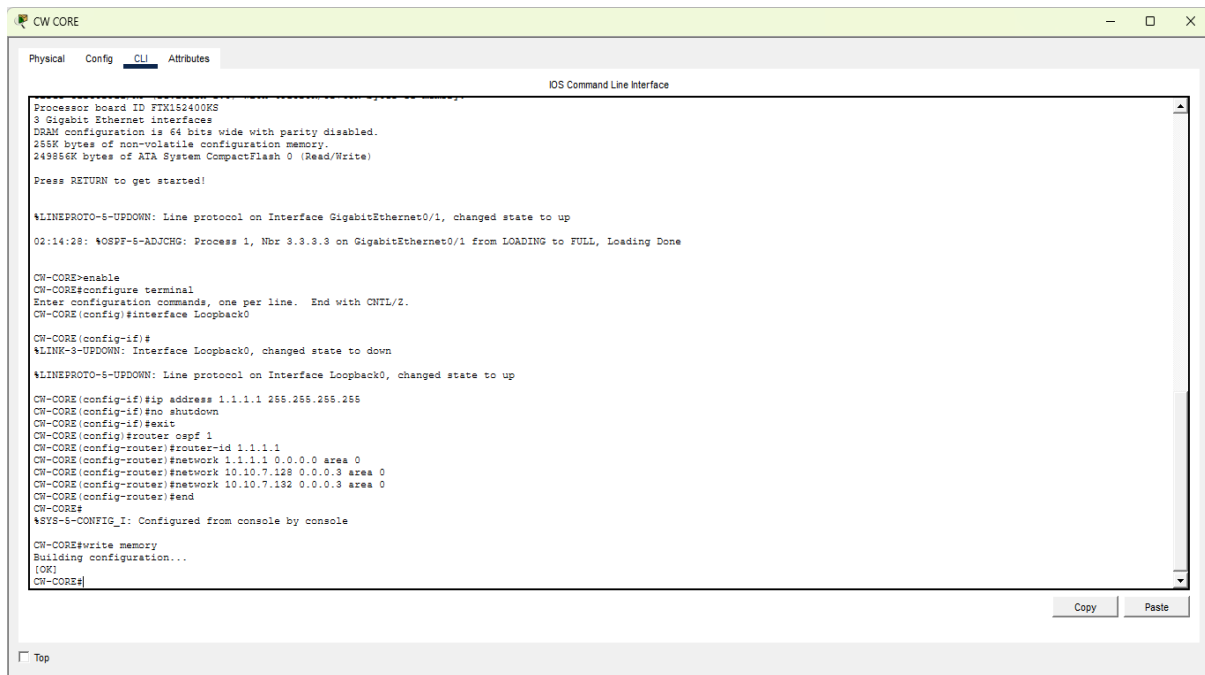
Ping statistics for 10.10.6.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 26ms, Average = 3ms

C:\>
```

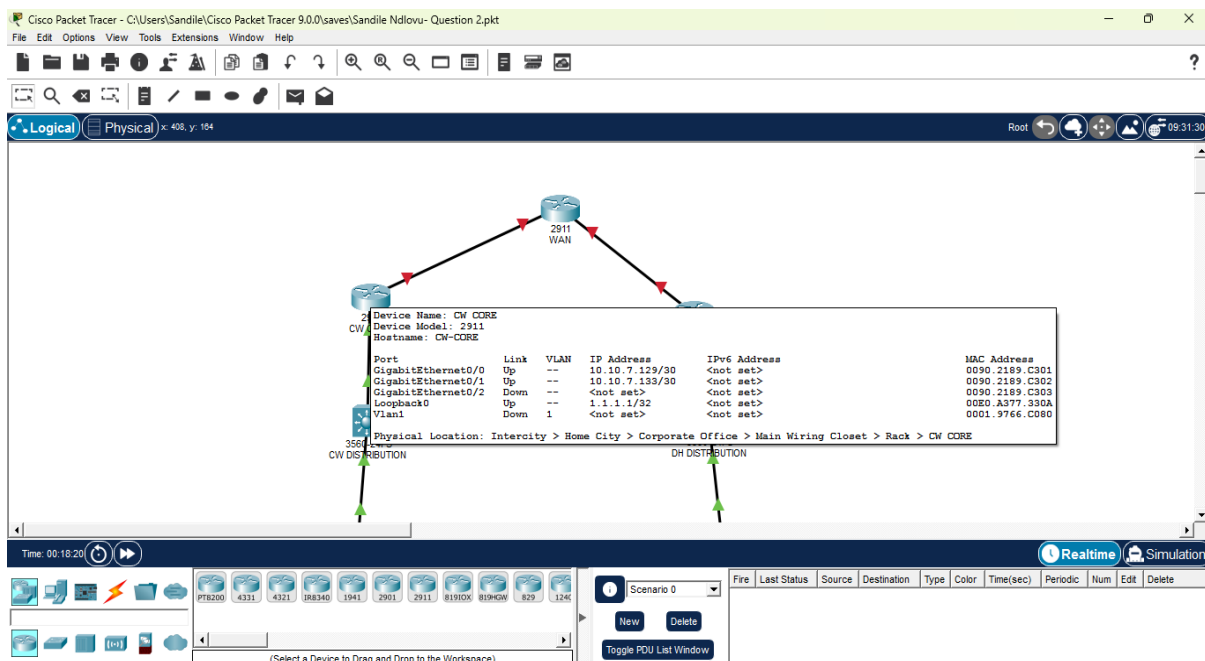
QUESTION 3

3.1) Configuration of OSPF AREA 0 for CW CORE AND CW DISTRIBUTION:

CW CORE: With the CW CORE we've configured it with Loopback0



Result of Loopback Configuration:



CW DISTRIBUTION: With the CW DISTRIBUTION we've configured it with the passive interface


```

CW DISTRIBUTION
Physical Config CLI Attributes
IOS Command Line Interface

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%LINK-6-CHANGED: Interface GigabitEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
%SPANTRIE-2-RECV_FVID_ERR: Received 902.1Q SPDU on non trunk GigabitEthernet0/2 VLAN1.
%SPANTRIE-2-BLOCK_FVID_LOCAL: Blocking GigabitEthernet0/2 on VLAN0001. Inconsistent port type.

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up
02:14:18: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on GigabitEthernet0/1 from LOADING to FULL, Loading Done

CW-DIST#enable
CW-DIST#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
CW-DIST(config)#router ospf 1
CW-DIST(config-router)#router-id 3.3.3.3
CW-DIST(config-router)#network 10.10.0.0 0.0.3.255 area 0
CW-DIST(config-router)#network 10.10.6.0 0.0.0.255 area 0
CW-DIST(config-router)#network 10.10.7.132 0.0.0.3 area 0
CW-DIST(config-router)#passive-interface vian 10
CW-DIST(config-router)#passive-interface vian 20
CW-DIST(config-router)#end
CW-DIST#
%SYS-5-CONFIG_I: Configured from console by console

CW-DIST#write memory
Building configuration...
[OK]
CW-DIST#

```

Result of passive interface configuration:

Port	Link	VLAN	IP Address	IPv6 Address	MAC Address
FastEthernet0/1	Down	1	<not set>	<not set>	000D.BDB7.DC01
FastEthernet0/2	Down	1	<not set>	<not set>	000D.BDB7.DC02
FastEthernet0/3	Down	1	<not set>	<not set>	000D.BDB7.DC03
FastEthernet0/4	Down	1	<not set>	<not set>	000D.BDB7.DC04
FastEthernet0/5	Down	1	<not set>	<not set>	000D.BDB7.DC05
FastEthernet0/6	Down	1	<not set>	<not set>	000D.BDB7.DC06
FastEthernet0/7	Down	1	<not set>	<not set>	000D.BDB7.DC07
FastEthernet0/8	Down	1	<not set>	<not set>	000D.BDB7.DC08
FastEthernet0/9	Down	1	<not set>	<not set>	000D.BDB7.DC09
FastEthernet0/10	Down	1	<not set>	<not set>	000D.BDB7.DC0A
FastEthernet0/11	Down	1	<not set>	<not set>	000D.BDB7.DC0B
FastEthernet0/12	Down	1	<not set>	<not set>	000D.BDB7.DC0C
FastEthernet0/13	Down	1	<not set>	<not set>	000D.BDB7.DC0D
FastEthernet0/14	Down	1	<not set>	<not set>	000D.BDB7.DC0E
FastEthernet0/15	Down	1	<not set>	<not set>	000D.BDB7.DC0F
FastEthernet0/16	Down	1	<not set>	<not set>	000D.BDB7.DC10
FastEthernet0/17	Down	1	<not set>	<not set>	000D.BDB7.DC11
FastEthernet0/18	Down	1	<not set>	<not set>	000D.BDB7.DC12
FastEthernet0/19	Down	1	<not set>	<not set>	000D.BDB7.DC13
FastEthernet0/20	Down	1	<not set>	<not set>	000D.BDB7.DC14
FastEthernet0/21	Down	1	<not set>	<not set>	000D.BDB7.DC15
FastEthernet0/22	Down	1	<not set>	<not set>	000D.BDB7.DC16
FastEthernet0/23	Down	1	<not set>	<not set>	000D.BDB7.DC17
FastEthernet0/24	Down	1	<not set>	<not set>	000D.BDB7.DC18
GigabitEthernet0/1	Up	1	10.10.7.134/30	<not set>	000D.BDB7.DC19
GigabitEthernet0/2	Up	--	<not set>	<not set>	000D.BDB7.DC1A
Vlan1	Down	1	<not set>	<not set>	0090.21B8.507B
Vlan10	Up	10	10.10.0.1/22	<not set>	0090.21B8.5001
Vlan20	Up	20	10.10.6.1/24	<not set>	0090.21B8.5002

Configuration of AREA 10 for DH CORE and DW DISTRIBUTION:

DH CORE: With the DH CORE we've configured it with Loopback0


```

DH DISTRIBUTION
Physical Config CLI Attributes
IOS Command Line Interface

%SPANTREE-2-BLOCK_PVID_LOCAL: Blocking GigabitEthernet0/2 on VLAN0001. Inconsistent port type.

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan30, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan40, changed state to up

DH-DIST#enable
DH-DIST#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
DH-DIST(config)#router ospf 1
DH-DIST(config-router)#router-id 4.4.4.4
DH-DIST(config-router)#network 10.10.4.0 0.0.1.255 area 10
DH-DIST(config-router)#
01:29:32: %OSPF-6-AREACHG: 10.10.4.0/0 changed from area 0 to area 10

DH-DIST(config-router)#network 10.10.7.0 0.0.0.127 area 10
DH-DIST(config-router)#
01:30:01: %OSPF-6-AREACHG: 10.10.7.0/0 changed from area 0 to area 10

DH-DIST(config-router)#network 10.10.7.136 0.0.0.3 area 10
DH-DIST(config-router)#
01:30:36: %OSPF-6-AREACHG: 10.10.7.136/0 changed from area 0 to area 10

DH-DIST(config-router)#passive-interface vlan 30
DH-DIST(config-router)#passive-interface vlan 40
DH-DIST(config-router)#end
DH-DIST#
%SYS-5-CONFIG_I: Configured from console by console

DH-DIST#write memory
Building configuration...
[OK]
DH-DIST#

```

Result of passive interface configuration:

Device Name: DH DISTRIBUTION
Device Model: 3560-24PS
Hostname: DH-DIST

Port	Link	VLAN	IP Address	IPv6 Address	MAC Address
FastEthernet0/1	Down	1	<not set>	<not set>	00E0.8F04.9001
FastEthernet0/2	Down	1	<not set>	<not set>	00E0.8F04.9002
FastEthernet0/3	Down	1	<not set>	<not set>	00E0.8F04.9003
FastEthernet0/4	Down	1	<not set>	<not set>	00E0.8F04.9004
FastEthernet0/5	Down	1	<not set>	<not set>	00E0.8F04.9005
FastEthernet0/6	Down	1	<not set>	<not set>	00E0.8F04.9006
FastEthernet0/7	Down	1	<not set>	<not set>	00E0.8F04.9007
FastEthernet0/8	Down	1	<not set>	<not set>	00E0.8F04.9008
FastEthernet0/9	Down	1	<not set>	<not set>	00E0.8F04.9009
FastEthernet0/10	Down	1	<not set>	<not set>	00E0.8F04.900A
FastEthernet0/11	Down	1	<not set>	<not set>	00E0.8F04.900B
FastEthernet0/12	Down	1	<not set>	<not set>	00E0.8F04.900C
FastEthernet0/13	Down	1	<not set>	<not set>	00E0.8F04.900D
FastEthernet0/14	Down	1	<not set>	<not set>	00E0.8F04.900E
FastEthernet0/15	Down	1	<not set>	<not set>	00E0.8F04.900F
FastEthernet0/16	Down	1	<not set>	<not set>	00E0.8F04.9010
FastEthernet0/17	Down	1	<not set>	<not set>	00E0.8F04.9011
FastEthernet0/18	Down	1	<not set>	<not set>	00E0.8F04.9012
FastEthernet0/19	Down	1	<not set>	<not set>	00E0.8F04.9013
FastEthernet0/20	Down	1	<not set>	<not set>	00E0.8F04.9014
FastEthernet0/21	Down	1	<not set>	<not set>	00E0.8F04.9015
FastEthernet0/22	Down	1	<not set>	<not set>	00E0.8F04.9016
FastEthernet0/23	Down	1	<not set>	<not set>	00E0.8F04.9017
FastEthernet0/24	Down	1	<not set>	<not set>	00E0.8F04.9018
GigabitEthernet0/1	Up	1	<not set>	<not set>	00E0.8F04.9019
GigabitEthernet0/2	Up	--	<not set>	<not set>	00E0.8F04.901A
Vlan1	Down	1	<not set>	<not set>	0002.1681.E76D
Vlan30	Up	30	10.10.4.1/23	<not set>	0002.1681.E701
Vlan40	Up	40	10.10.7.136/25	<not set>	0002.1681.E702

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > DH DISTRIBUTION

3560-24PS CW DISTRIBUTION

3560-24PS DH DISTRIBUTION

Time: 00:19:40

Realtime Simulation

Scenario 0

Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit Delete

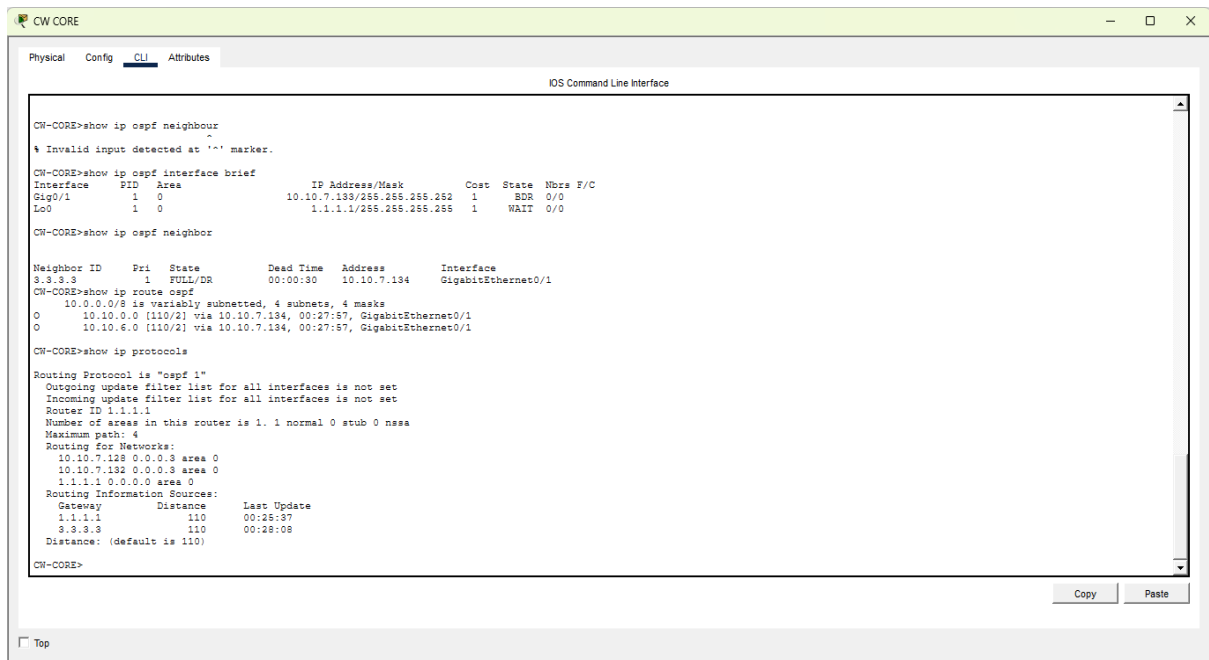
New Delete

Toggle PDU List Window

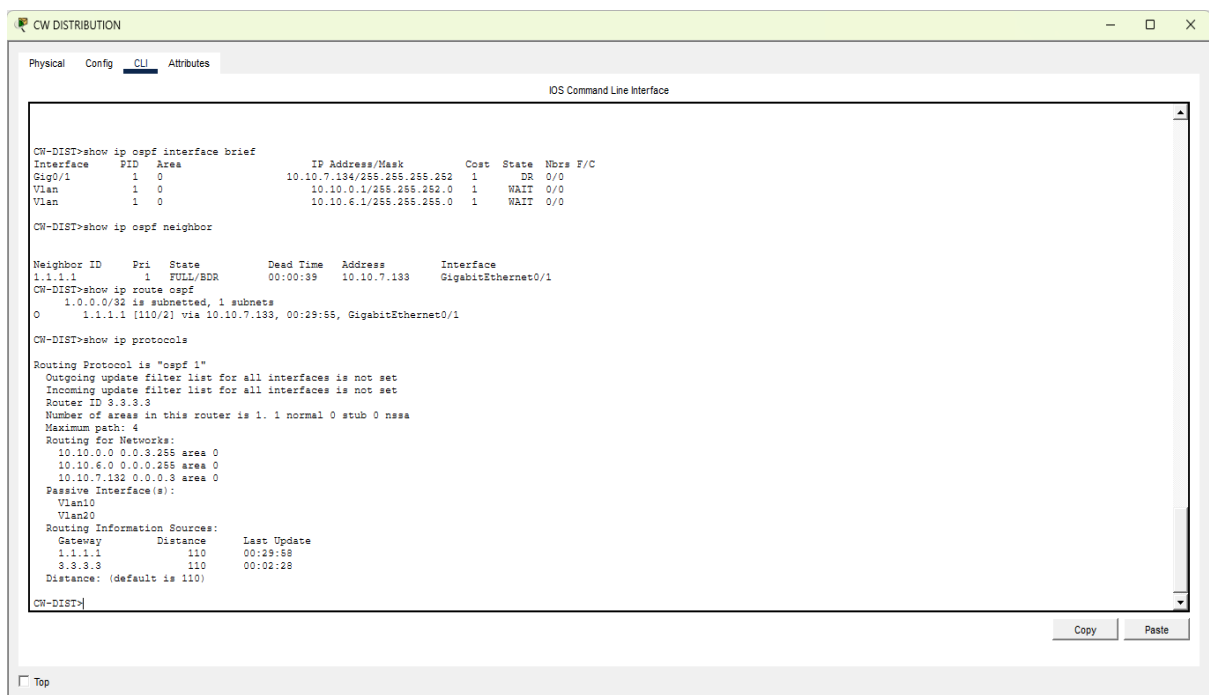
(Select a Device to Drag and Drop to the Workspace)

CW CORE and CW DISTRIBUTION worked after putting in the command 'show ip ospf neighbor'

CW CORE:

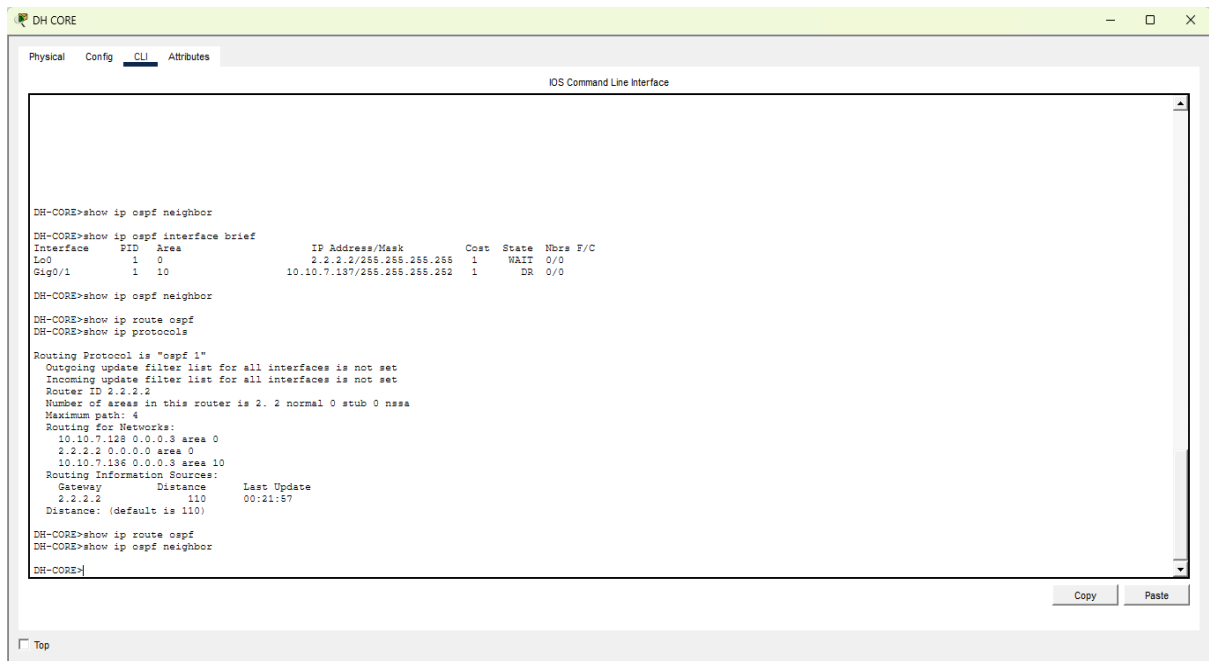


CW DISTRIBUTION:



DH CORE and DH DISTRIBUTION showed results after putting in the command 'show IP route'

DH CORE:



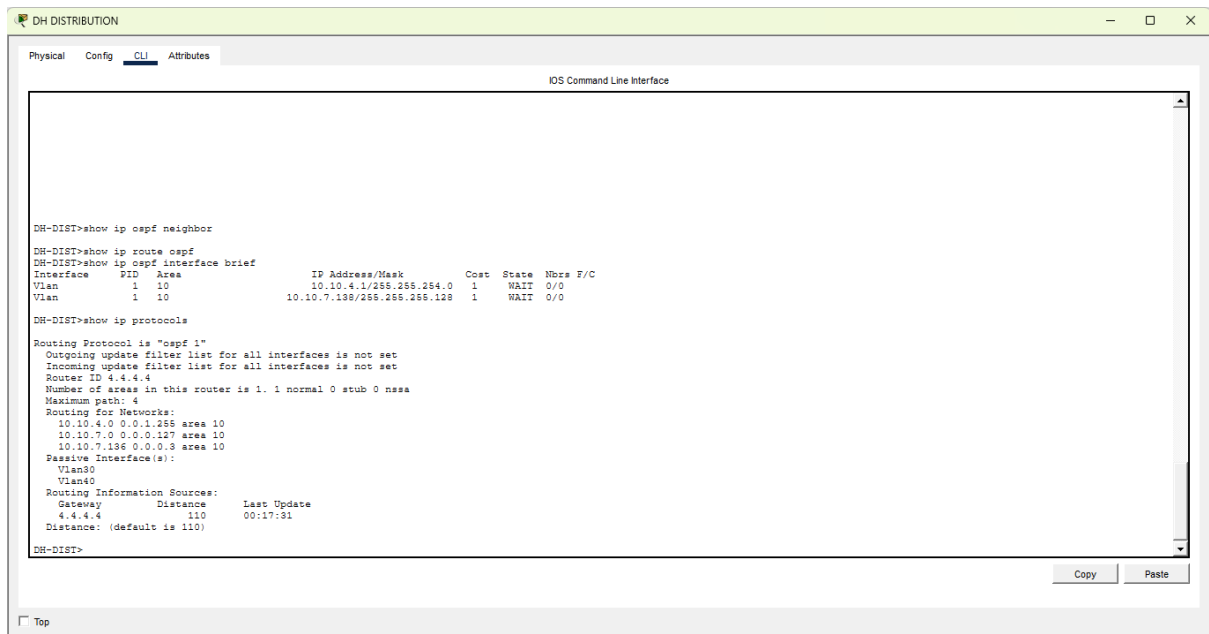
The screenshot shows the CLI of a router named 'DH CORE'. The user has entered several commands to check the OSPF configuration and status. The output shows that OSPF is running on interfaces 'Lo0' and 'Gig0/1'. The router ID is 2.2.2.2, and the network is 10.10.7.137/255.255.255.252. The status of the interfaces is 'DR' (Designated Router) for 'Gig0/1' and 'WAIT' for 'Lo0'.

```
DH-CORE>show ip ospf neighbor
DH-CORE>show ip ospf interface brief
DH-CORE>show ip route ospf
DH-CORE>show ip protocols

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 2.2.2.2
  Number of areas in this router is 2. 2 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.10.7.138 0.0.0.0 area 0
    2.2.2.2 0.0.0.0 area 0
    10.10.7.136 0.0.0.0 area 10
  Routing Information Sources:
    Gateway         Distance      Last Update
    2.2.2.2          110          00:21:57
  Distance: (default is 110)

DH-CORE>show ip route ospf
DH-CORE>show ip ospf neighbor
DH-CORE>
```

DH DISTRIBUTION:



The screenshot shows the CLI of a router named 'DH DISTRIBUTION'. The user has entered several commands to check the OSPF configuration and status. The output shows that OSPF is running on interfaces 'Vlan10' and 'Vlan40'. The router ID is 4.4.4.4, and the network is 10.10.4.1/255.255.254.0. The status of the interfaces is 'WAIT' (Waiting for Designated Router) for both 'Vlan10' and 'Vlan40'.

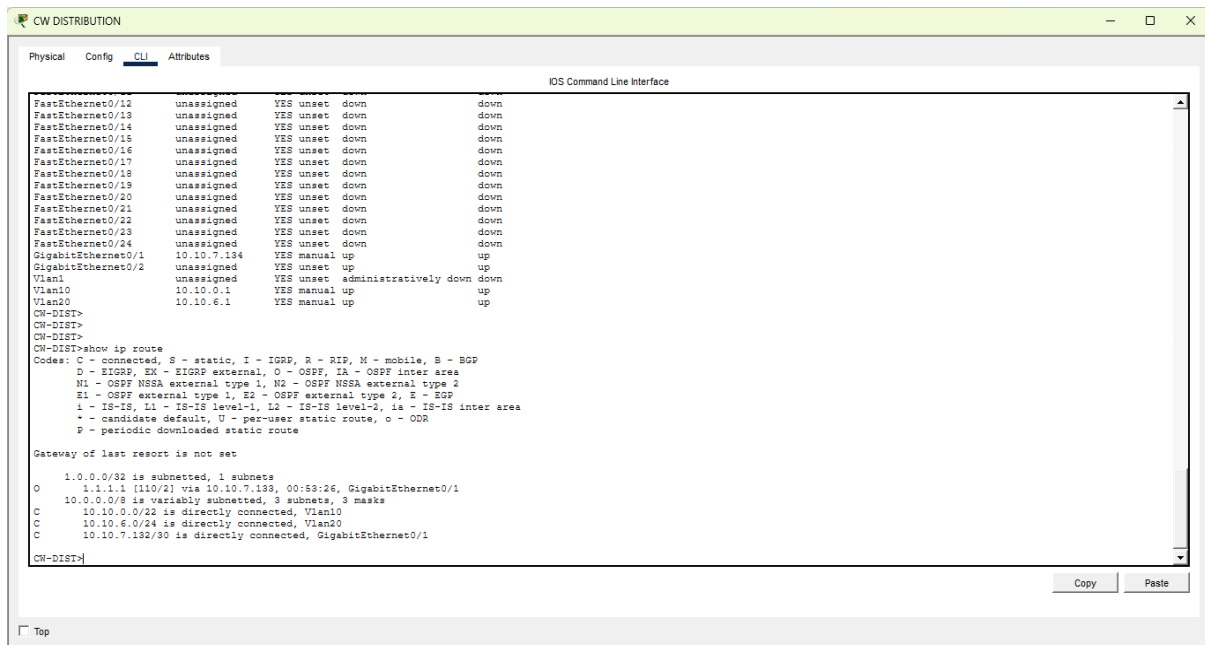
```
DH-DIST>show ip ospf neighbor
DH-DIST>show ip route ospf
DH-DIST>show ip ospf interface brief
DH-DIST>show ip protocols

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 4.4.4.4
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.10.4.0 0.0.1.255 area 10
    10.10.7.0 0.0.0.127 area 10
    10.10.7.136 0.0.0.0 area 10
  Passive Interface(s):
    Vlan30
    Vlan40
  Routing Information Sources:
    Gateway         Distance      Last Update
    4.4.4.4          110          00:17:31
  Distance: (default is 110)

DH-DIST>
```

3.2) With the Layer 3 switches – CW DISTRIBUTION and DH DISTRIBUTION we've configured them to show that all the VLAN's under each respective have their own gateway. We've also confirmed that the VLAN's are communicating and the connectivity is showing.

CW DISTRIBUTION:



The screenshot shows the CW DISTRIBUTION configuration window with the CLI tab selected. The output of the 'show ip route' command is displayed, showing the routing table for the device. The table lists various interfaces and their status, including FastEthernet0/12 through FastEthernet0/24, GigabitEthernet0/1, and GigabitEthernet0/2. The output also shows the configuration of VLANs 10, 20, and 30, and the status of the CW-DIST interface.

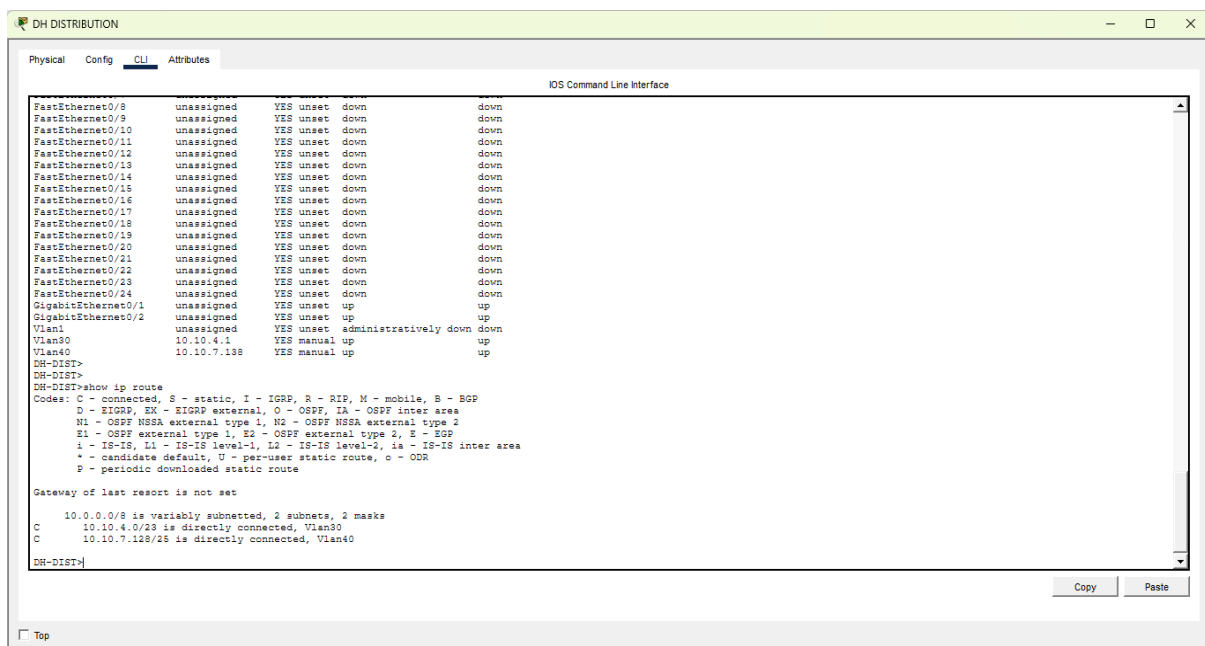
```
Physical Config CLI Attributes
IOS Command Line Interface

FastEthernet0/12 unassigned YES unset down down
FastEthernet0/13 unassigned YES unset down down
FastEthernet0/14 unassigned YES unset down down
FastEthernet0/15 unassigned YES unset down down
FastEthernet0/16 unassigned YES unset down down
FastEthernet0/17 unassigned YES unset down down
FastEthernet0/18 unassigned YES unset down down
FastEthernet0/19 unassigned YES unset down down
FastEthernet0/20 unassigned YES unset down down
FastEthernet0/21 unassigned YES unset down down
FastEthernet0/22 unassigned YES unset down down
FastEthernet0/23 unassigned YES unset down down
FastEthernet0/24 unassigned YES unset down down
GigabitEthernet0/1 10.10.7.134 YES manual up up
GigabitEthernet0/2 unassigned YES unset up up
Vlan1 unassigned YES administratively down down
Vlan10 10.10.0.1 YES manual up up
Vlan20 10.10.0.1 YES manual up up
CW-DIST>
CW-DIST>
CW-DIST>show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       I - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/32 is subnetted, 1 subnets
O 10.0.0.0/32 is directly connected, GigabitEthernet0/1
O 10.0.0.0/8 is variably subnetted, 3 subnets, 3 masks
C 10.10.0.0/22 is directly connected, Vlan10
C 10.10.6.0/24 is directly connected, Vlan20
C 10.10.7.132/30 is directly connected, GigabitEthernet0/1
CW-DIST>
```

DH DISTRIBUTION:



The screenshot shows the DH DISTRIBUTION configuration window with the CLI tab selected. The output of the 'show ip route' command is displayed, showing the routing table for the device. The table lists various interfaces and their status, including FastEthernet0/8 through FastEthernet0/24, GigabitEthernet0/1, and GigabitEthernet0/2. The output also shows the configuration of VLANs 30, 40, and 60, and the status of the DH-DIST interface.

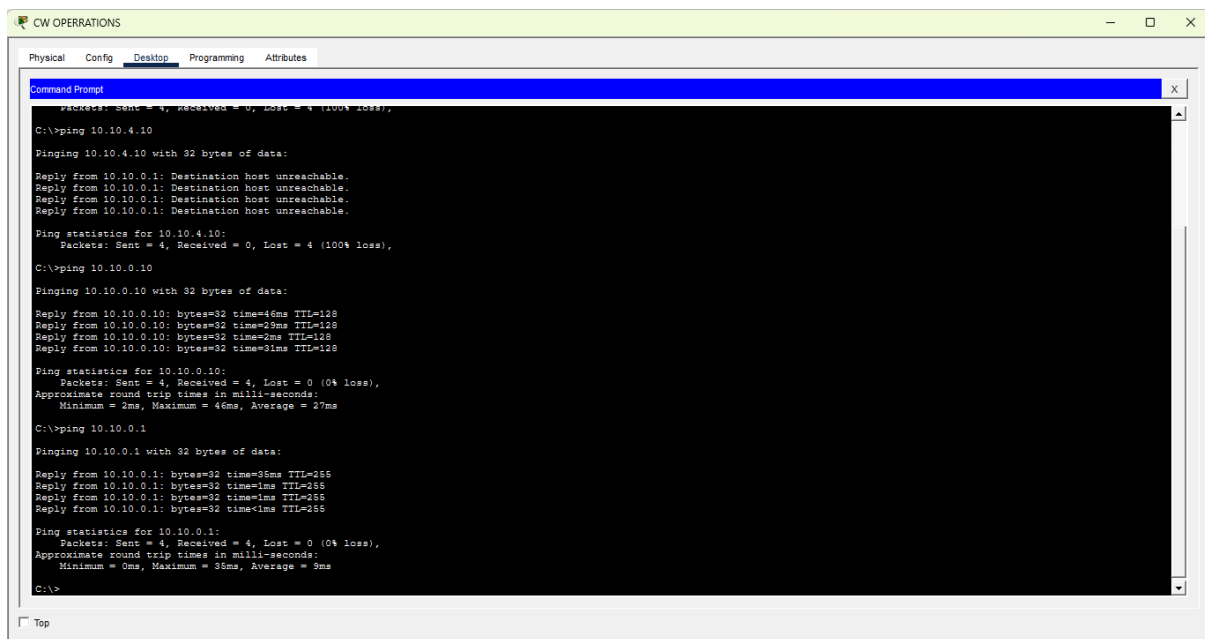
```
Physical Config CLI Attributes
IOS Command Line Interface

FastEthernet0/8 unassigned YES unset down down
FastEthernet0/9 unassigned YES unset down down
FastEthernet0/10 unassigned YES unset down down
FastEthernet0/11 unassigned YES unset down down
FastEthernet0/12 unassigned YES unset down down
FastEthernet0/13 unassigned YES unset down down
FastEthernet0/14 unassigned YES unset down down
FastEthernet0/15 unassigned YES unset down down
FastEthernet0/16 unassigned YES unset down down
FastEthernet0/17 unassigned YES unset down down
FastEthernet0/18 unassigned YES unset down down
FastEthernet0/19 unassigned YES unset down down
FastEthernet0/20 unassigned YES unset down down
FastEthernet0/21 unassigned YES unset down down
FastEthernet0/22 unassigned YES unset down down
FastEthernet0/23 unassigned YES unset down down
FastEthernet0/24 unassigned YES unset down down
GigabitEthernet0/1 unassigned YES unset up up
GigabitEthernet0/2 unassigned YES unset administratively down down
Vlan1 unassigned YES administratively down down
Vlan30 10.10.4.1 YES manual up up
Vlan40 10.10.7.138 YES manual up up
DH-DIST>
DH-DIST>
DH-DIST>show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       I - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 10.10.4.0/23 is directly connected, Vlan30
C 10.10.7.128/26 is directly connected, Vlan40
DH-DIST>
```

Connectivity between devices and VLAN's



```
CW OPERATIONS
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ping 10.10.4.10
Pinging 10.10.4.10 with 32 bytes of data:
Reply from 10.10.0.1: Destination host unreachable.
Reply from 10.10.0.1: Destination host unreachable.
Reply from 10.10.0.1: Destination host unreachable.
Reply from 10.10.0.1: Destination host unreachable.
Ping statistics for 10.10.4.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.10.0.10
Pinging 10.10.0.10 with 32 bytes of data:
Reply from 10.10.0.10: bytes=32 time=46ms TTL=128
Reply from 10.10.0.10: bytes=32 time=29ms TTL=128
Reply from 10.10.0.10: bytes=32 time=2ms TTL=128
Reply from 10.10.0.10: bytes=32 time=31ms TTL=128
Ping statistics for 10.10.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 46ms, Average = 27ms

C:\>ping 10.10.0.1
Pinging 10.10.0.1 with 32 bytes of data:
Reply from 10.10.0.1: bytes=32 time=35ms TTL=255
Reply from 10.10.0.1: bytes=32 time=1ms TTL=255
Reply from 10.10.0.1: bytes=32 time=1ms TTL=255
Reply from 10.10.0.1: bytes=32 time=1ms TTL=255
Ping statistics for 10.10.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 35ms, Average = 9ms

C:\>
```

3.1d)

The advantage of using a Layer 3 SVI rather than a Router on a stick is because a layer 3 switch with an SVI is more efficient because it can perform inter-VLAN routing at a high speed.

SVIs also allow multiple VLANs to have their own default gateways on the same multilayer switch. This helps reduce dependency on a single router trunk. This also helps provide better scalability and performance.

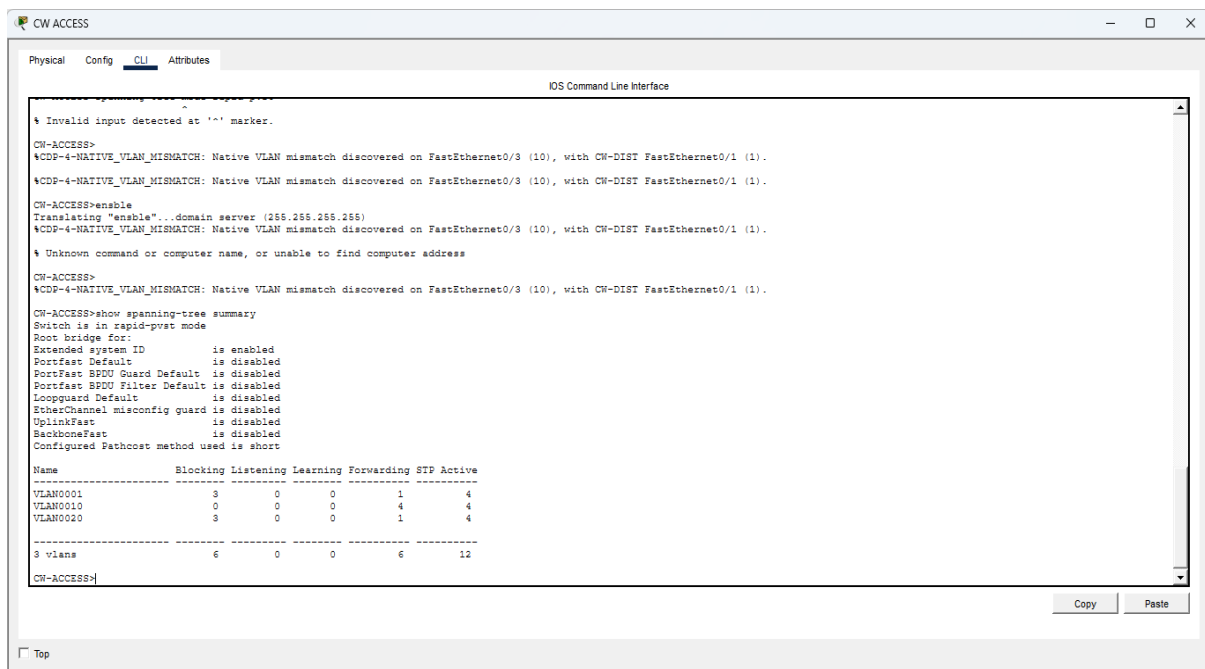
QUESTION 4

- *Disclaimer: It's come to my attention that when it came to configuring for Q4, it was supposed to be with GigabitEthernet for the second connection of the layer 3 and layer 2 switches. While the first connection does have GigabitEthernet the second connection only provided FastEthernet connection. I hope it doesn't affect the marking.*

4.1a)

Configuration of the Rapid Spanning tree protocol for CW:

CW ACCESS:



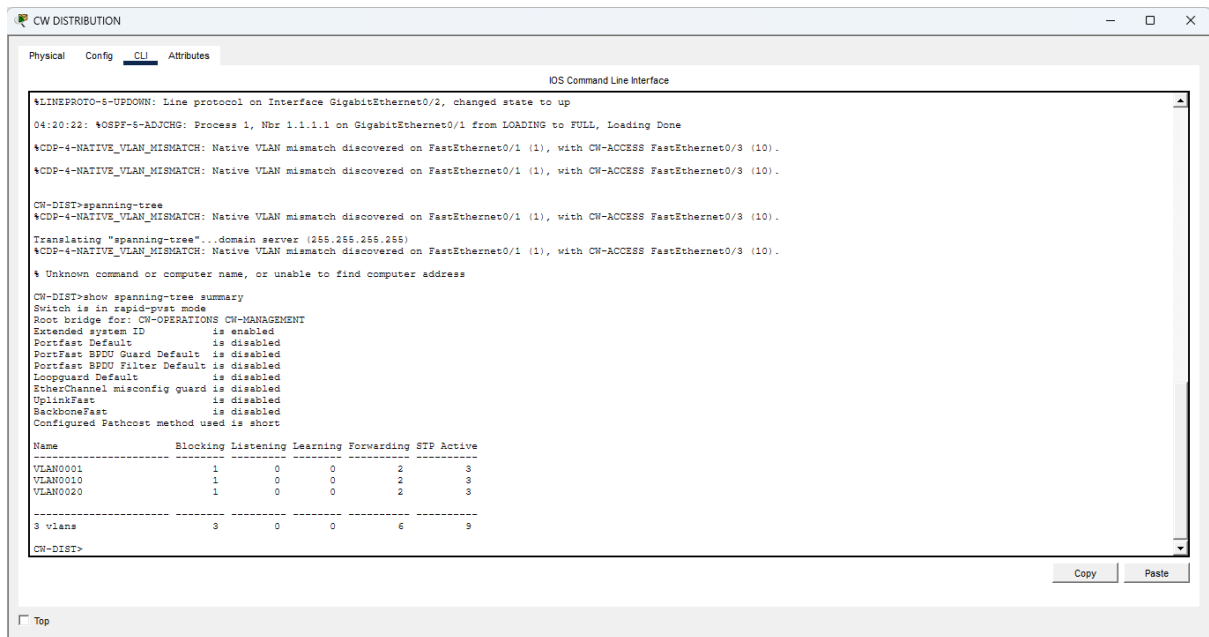
```
Invalid input detected at '^' marker.
CW-ACCESS>
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/3 (10), with CW-DIST FastEthernet0/1 (1).
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/3 (10), with CW-DIST FastEthernet0/1 (1).
CW-ACCESS>enable
Translating "enable"...domain server (255.255.255.255)
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/3 (10), with CW-DIST FastEthernet0/1 (1).
% Unknown command or computer name, or unable to find computer address
CW-ACCESS>
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/3 (10), with CW-DIST FastEthernet0/1 (1).
CW-ACCESS>show spanning-tree summary
Switch is in rapid-pvst mode
Root bridge for:
Extended system ID is enabled
Portfast Default is disabled
Portfast BPDU Guard Default is disabled
Portfast BPDU Filter Default is disabled
Loopguard Default is disabled
EtherChannel misconfig guard is disabled
UplinkFast is disabled
BackboneFast is disabled
Configured Pathcost method used is short

```

Name	Blocking	Listening	Learning	Forwarding	STP Active
VLAN0001	3	0	0	1	4
VLAN0010	0	0	0	4	4
VLAN0020	3	0	0	1	4
3 vlans	6	0	0	6	12

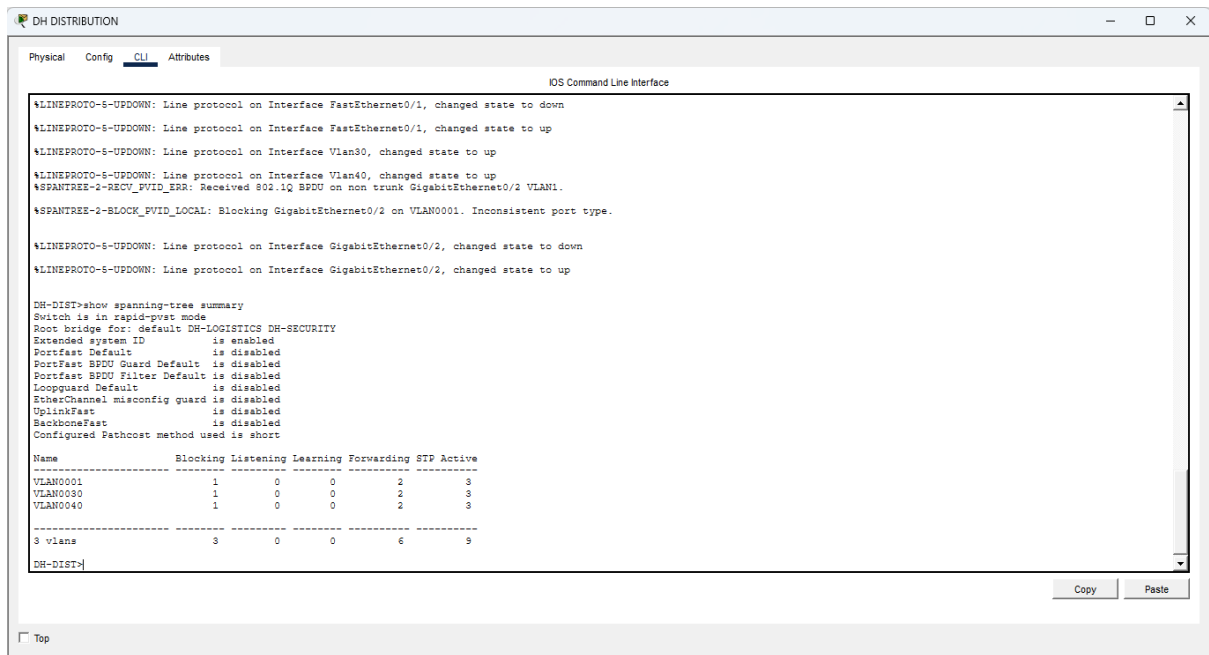
```
CW-ACCESS#
```

CW DISTRIBUTION:

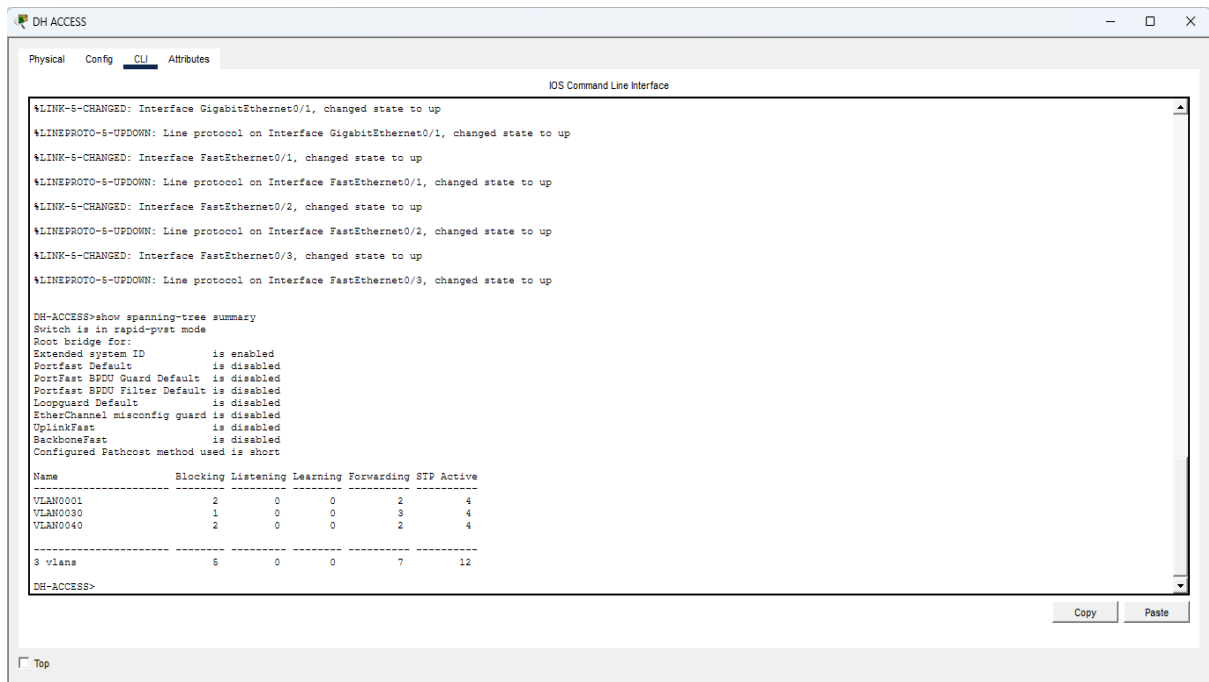


Configuration of Rapid spanning tree protocol for DH:

DH DISTRIBUTION:

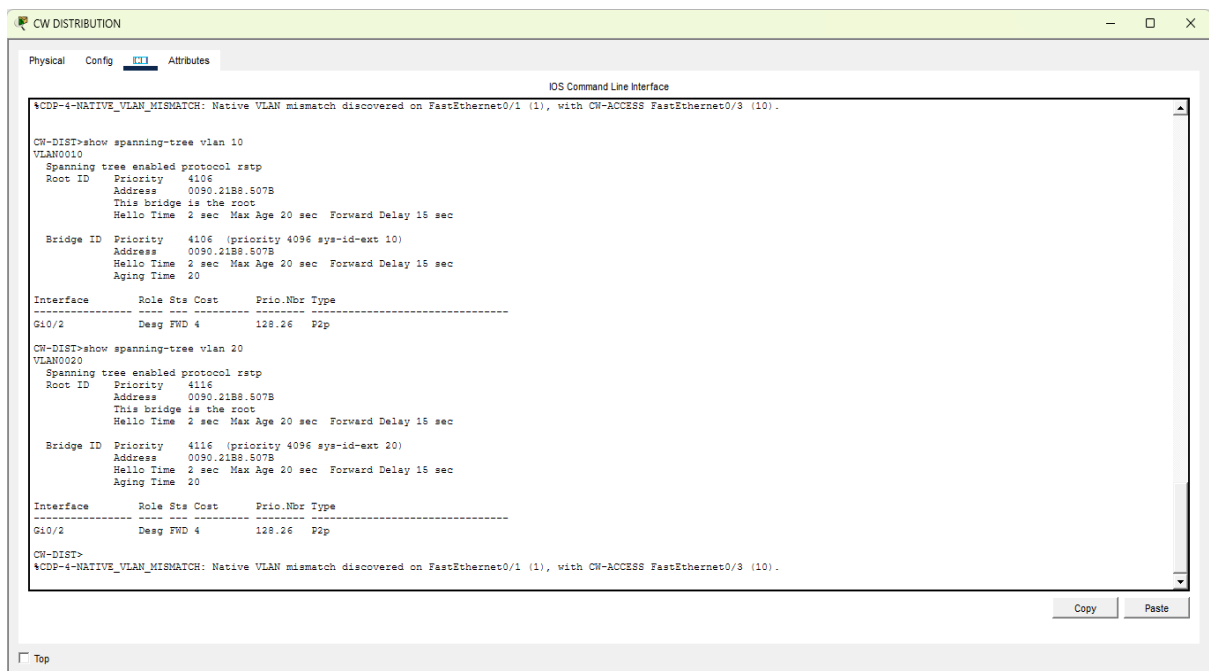


DH ACCESS:

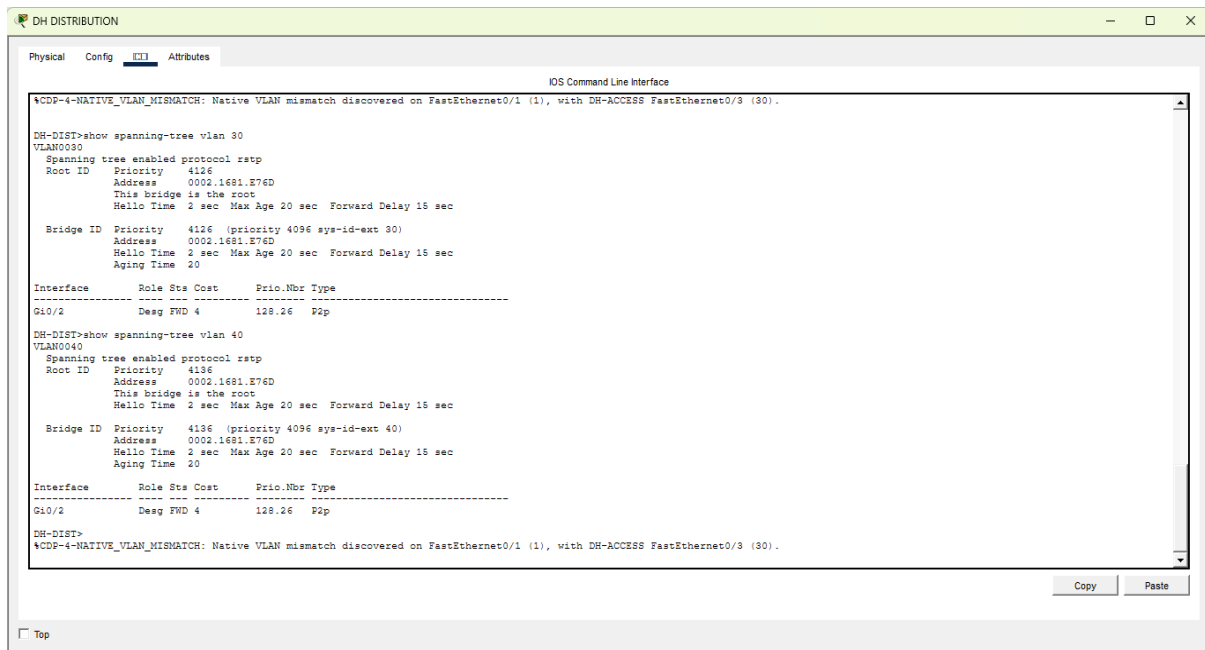


4.1b)

Root bridge for CW DISTRIBUTION;



Root bridge for DH DISTRIBUTION:



The screenshot shows a network configuration window titled "DH DISTRIBUTION" with tabs for Physical, Config, CLI, and Attributes. The CLI tab is active, displaying the IOS Command Line Interface. The configuration shows two VLANs, 30 and 40, both with a root bridge ID of 4126 and address 0002.1681.E76D. The root bridge is configured with a priority of 4126 and a sys-id-ext of 30. The bridge ID is 4126 (priority 4096 sys-id-ext 30). The bridge is configured with a hello time of 2 sec, max age of 20 sec, and forward delay of 15 sec. The bridge is also configured with an aging time of 20. The bridge is connected to interface Gi0/2, which is configured with a role of Desg FWD 4, a priority of 128.26, and a type of P2p.

```
$CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with DH-ACCESS FastEthernet0/3 (30).

DH-DIST#show spanning-tree vlan 30
VLAN0030
  Spanning tree enabled protocol rstp
  Root ID    Priority    4126
             Address     0002.1681.E76D
             This bridge is the root
             Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    4126 (priority 4096 sys-id-ext 30)
             Address     0002.1681.E76D
             Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface                Role Sts Cost      Prio.Nbr Type
-----
Gi0/2                    Desg FWD 4      128.26  P2p

DH-DIST#show spanning-tree vlan 40
VLAN0040
  Spanning tree enabled protocol rstp
  Root ID    Priority    4136
             Address     0002.1681.E76D
             This bridge is the root
             Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec

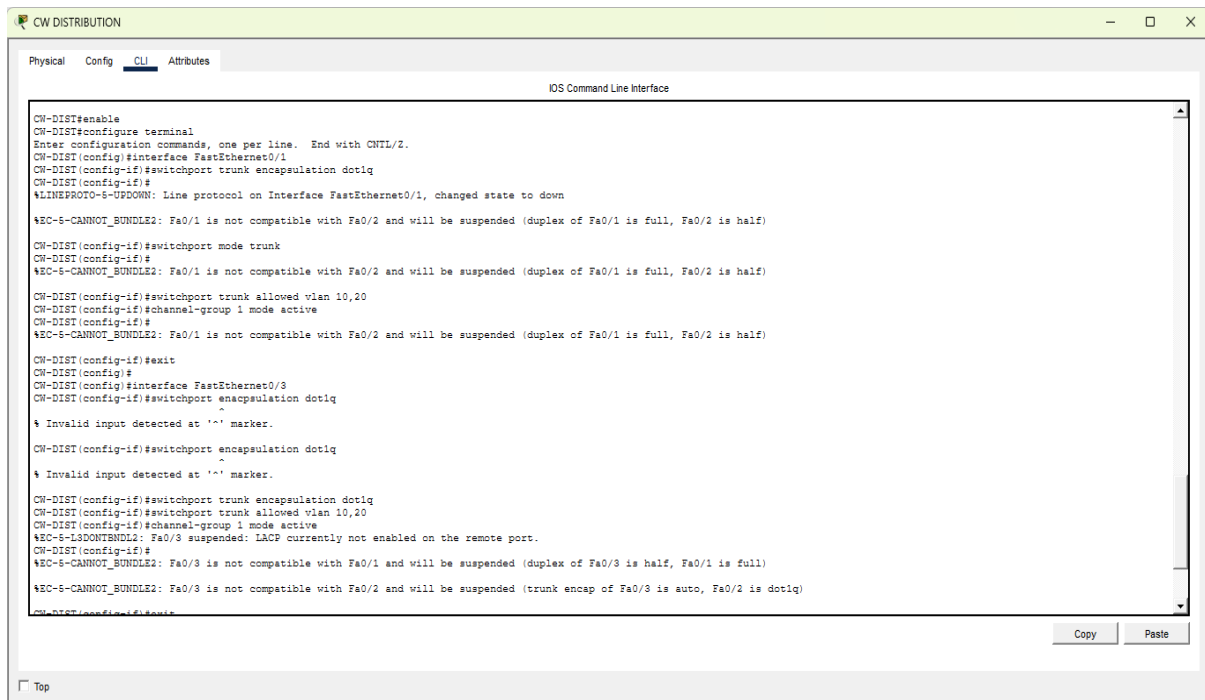
  Bridge ID  Priority    4136 (priority 4096 sys-id-ext 40)
             Address     0002.1681.E76D
             Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface                Role Sts Cost      Prio.Nbr Type
-----
Gi0/2                    Desg FWD 4      128.26  P2p

DH-DIST#
$CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with DH-ACCESS FastEthernet0/3 (30).
```

4.1c/d) Configuration of the EtherChannel for CW:

CW DISTRIBUTION:



The screenshot shows a network configuration window titled "CW DISTRIBUTION" with tabs for Physical, Config, CLI, and Attributes. The CLI tab is active, displaying the IOS Command Line Interface. The configuration shows the setup of an EtherChannel for a spanning tree protocol. The root bridge is configured with a priority of 4126 and a sys-id-ext of 30. The bridge ID is 4126 (priority 4096 sys-id-ext 30). The bridge is configured with a hello time of 2 sec, max age of 20 sec, and forward delay of 15 sec. The bridge is also configured with an aging time of 20. The bridge is connected to interface Gi0/2, which is configured with a role of Desg FWD 4, a priority of 128.26, and a type of P2p.

```
CW-DIST#enable
CW-DIST#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
CW-DIST(config)#interface FastEthernet0/1
CW-DIST(config-if)#switchport trunk encapsulation dot1q
CW-DIST(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%EC-5-CANNOT_BUNDLE2: Fa0/1 is not compatible with Fa0/2 and will be suspended (duplex of Fa0/1 is full, Fa0/2 is half)
CW-DIST(config-if)#switchport mode trunk
CW-DIST(config-if)#
%EC-5-CANNOT_BUNDLE2: Fa0/1 is not compatible with Fa0/2 and will be suspended (duplex of Fa0/1 is full, Fa0/2 is half)
CW-DIST(config-if)#switchport trunk allowed vlan 10,20
CW-DIST(config-if)#channel-group 1 mode active
CW-DIST(config-if)#
%EC-5-CANNOT_BUNDLE2: Fa0/1 is not compatible with Fa0/2 and will be suspended (duplex of Fa0/1 is full, Fa0/2 is half)
CW-DIST(config-if)#exit
CW-DIST(config)#
CW-DIST(config)#interface FastEthernet0/3
CW-DIST(config-if)#switchport encapsulation dot1q
CW-DIST(config-if)#
% Invalid input detected at '^' marker.
CW-DIST(config-if)#switchport encapsulation dot1q
CW-DIST(config-if)#
% Invalid input detected at '^' marker.
CW-DIST(config-if)#switchport trunk encapsulation dot1q
CW-DIST(config-if)#switchport trunk allowed vlan 10,20
CW-DIST(config-if)#channel-group 1 mode active
%EC-5-L3DONTBUNDLE2: Fa0/3 suspended: LACP currently not enabled on the remote port.
CW-DIST(config-if)#
%EC-5-CANNOT_BUNDLE2: Fa0/3 is not compatible with Fa0/1 and will be suspended (duplex of Fa0/3 is half, Fa0/1 is full)
%EC-5-CANNOT_BUNDLE2: Fa0/3 is not compatible with Fa0/2 and will be suspended (trunk encap of Fa0/3 is auto, Fa0/2 is dot1q)
CW-DIST(config-if)#exit
```

```
CW-DIST(config-if)#
%EC-5-CANNOT_BUNDLE2: Fa0/1 is not compatible with Fa0/2 and will be suspended (duplex of Fa0/1 is full, Fa0/2 is half)
CW-DIST(config-if)#switchport trunk allowed vlan 10,20
CW-DIST(config-if)#channel-group 1 mode active
CW-DIST(config-if)#
%EC-5-CANNOT_BUNDLE2: Fa0/1 is not compatible with Fa0/2 and will be suspended (duplex of Fa0/1 is full, Fa0/2 is half)
CW-DIST(config-if)#exit
CW-DIST(config)#
CW-DIST(config)#interface FastEthernet0/3
CW-DIST(config-if)#switchport encapsulation dot1q
^
% Invalid input detected at '^' marker.
CW-DIST(config-if)#switchport encapsulation dot1q
^
% Invalid input detected at '^' marker.
CW-DIST(config-if)#switchport trunk encapsulation dot1q
CW-DIST(config-if)#switchport trunk allowed vlan 10,20
CW-DIST(config-if)#channel-group 1 mode active
%EC-5-1SDONTBUNDLE2: Fa0/3 suspended: LACP currently not enabled on the remote port.
CW-DIST(config-if)#
%EC-5-CANNOT_BUNDLE2: Fa0/3 is not compatible with Fa0/1 and will be suspended (duplex of Fa0/3 is half, Fa0/1 is full)
%EC-5-CANNOT_BUNDLE2: Fa0/3 is not compatible with Fa0/2 and will be suspended (trunk encap of Fa0/3 is auto, Fa0/2 is dot1q)
CW-DIST(config-if)#exit
CW-DIST(config)#interface Port-channel
CW-DIST(config-if)#switchport trunk encapsulation dot1q
CW-DIST(config-if)#switchport trunk allowed vlan 10,20
CW-DIST(config-if)#end
CW-DIST#
%SYS-8-CONFIG_I: Configured from console by console
CW-DIST#write memory
Building configuration...
[OK]
CW-DIST#
```

Result of EtherChannel Configuration;

```
CW-DIST con0 is now available

Press RETURN to get started.

CW-DIST>show etherchannel summary
Flags: D - down        P - in port-channel
       T - stand-alone  S - suspended
       H - Hot-standby (LACP only)
       R - Layer3       S - Layer2
       U - in use       f - failed to allocate aggregator
       u - unsuitable for bundling
       w - waiting to be aggregated
       d - default port

Number of channel-groups in use: 1
Number of aggregators:          1

Group  Port-channel  Protocol    Ports
-----
1      Po1(SD)          LACP       Fa0/1(s) Fa0/2(s) Fa0/3(s)
CW-DIST#
```

CW ACCESS:

The screenshot shows the CW ACCESS CLI interface with the following commands and output:

```
! Incomplete command.
CW-ACCESS(config)#exit
CW-ACCESS#
%SYS-5-CONFIG_I: Configured from console by console

CW-ACCESS#enable
CW-ACCESS#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
CW-ACCESS(config)#interface FastEthernet0/1
CW-ACCESS(config-if)#switchport trunk allowed vlan 10,20
CW-ACCESS(config-if)#channel-group 1 mode active
%EC-5-L3DONTBNDL2: Fa0/1 suspended: LACP currently not enabled on the remote port.
CW-ACCESS(config-if)#
Creating a port-channel interface Port-channel 1

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

CW-ACCESS(config-if)#exit
CW-ACCESS(config)#interface FastEthernet0/3
CW-ACCESS(config-if)#switchport trunk allowed vlan 10,20
CW-ACCESS(config-if)#channel-group 1 mode active
%EC-5-L3DONTBNDL2: Fa0/3 suspended: LACP currently not enabled on the remote port.
CW-ACCESS(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up

CW-ACCESS(config-if)#exit
CW-ACCESS(config)#interface Port-channel1
CW-ACCESS(config-if)#switchport trunk allowed vlan 10,20
CW-ACCESS(config-if)#end
CW-ACCESS#
%SYS-5-CONFIG_I: Configured from console by console

CW-ACCESS#write memory
Building configuration...
[OK]
CW-ACCESS#
```

Result of EtherChannel Configuration:

The screenshot shows the CW ACCESS CLI interface with the following commands and output:

```
CW-ACCESS con0 is now available

Press RETURN to get started.

CW-ACCESS>show etherchannel summary
Flags: D - down        P - in port-channel
       I - stand-alone s - suspended
       H - Hot-standby (LACP only)
       A - Layer3      S - Layer2
       U - in use      f - failed to allocate aggregator
       u - unsuitable for bundling
       W - waiting to be aggregated
       d - default port

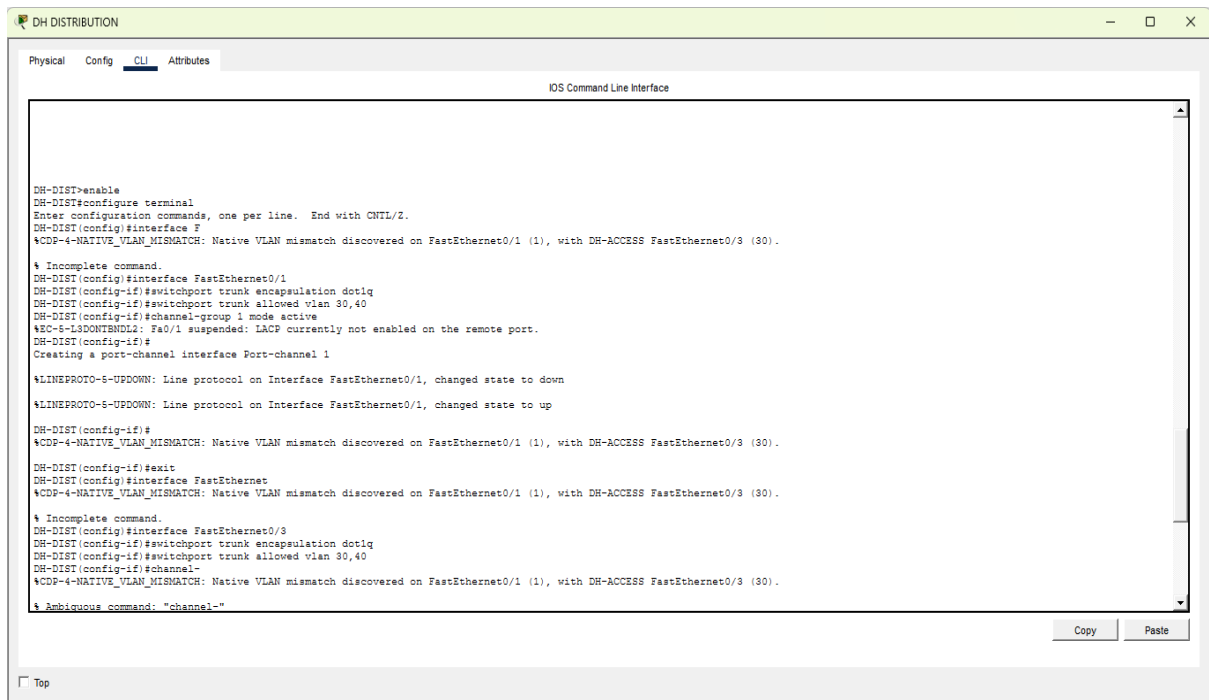
Number of channel-groups in use: 1
Number of aggregators:          1

Group  Port-channel  Protocol    Ports
-----
1      Po1(SD)          LACP       Fa0/1(I) Fa0/3(I)

CW-ACCESS>
```

Configuration of the EtherChannel for DH:

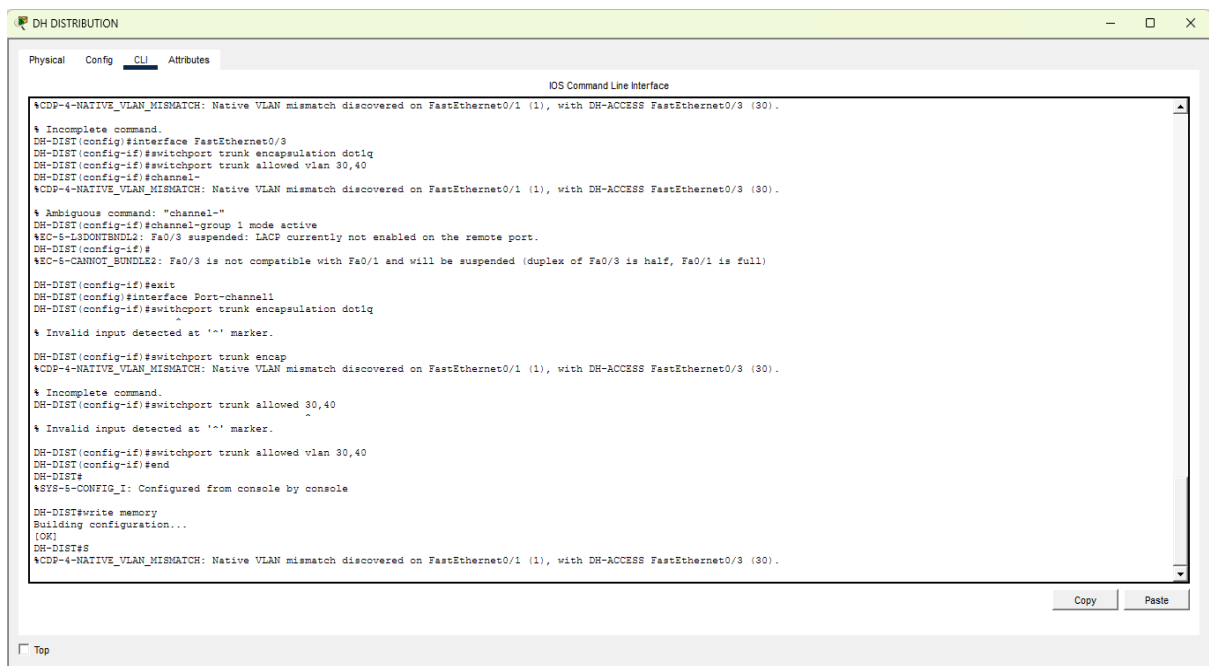
DH DISTRIBUTION:



The screenshot shows a terminal window titled "DH DISTRIBUTION" with tabs for Physical, Config, CLI, and Attributes. The CLI tab is active, displaying the "IOS Command Line Interface". The terminal output shows the following commands and responses:

```
DH-DIST>enable
DH-DIST#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
DH-DIST(config)#interface F
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with DH-ACCESS FastEthernet0/3 (30).
% Incomplete command.
DH-DIST(config)#interface FastEthernet0/1
DH-DIST(config-if)#switchport trunk encapsulation dot1q
DH-DIST(config-if)#switchport trunk allowed vlan 30,40
DH-DIST(config-if)#channel-group 1 mode active
%EC-5-L3CONTBNDL2: Fa0/1 suspended: LACP currently not enabled on the remote port.
DH-DIST(config-if)#
Creating a port-channel interface Port-channel 1
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
DH-DIST(config-if)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with DH-ACCESS FastEthernet0/3 (30).
DH-DIST(config-if)#exit
DH-DIST(config)#interface FastEthernet
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with DH-ACCESS FastEthernet0/3 (30).
% Incomplete command.
DH-DIST(config)#interface FastEthernet0/3
DH-DIST(config-if)#switchport trunk encapsulation dot1q
DH-DIST(config-if)#switchport trunk allowed vlan 30,40
DH-DIST(config-if)#channel-
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with DH-ACCESS FastEthernet0/3 (30).
% Ambiguous command: "channel-"
```

At the bottom right of the terminal window, there are "Copy" and "Paste" buttons. A "Top" link is located at the bottom left of the window.

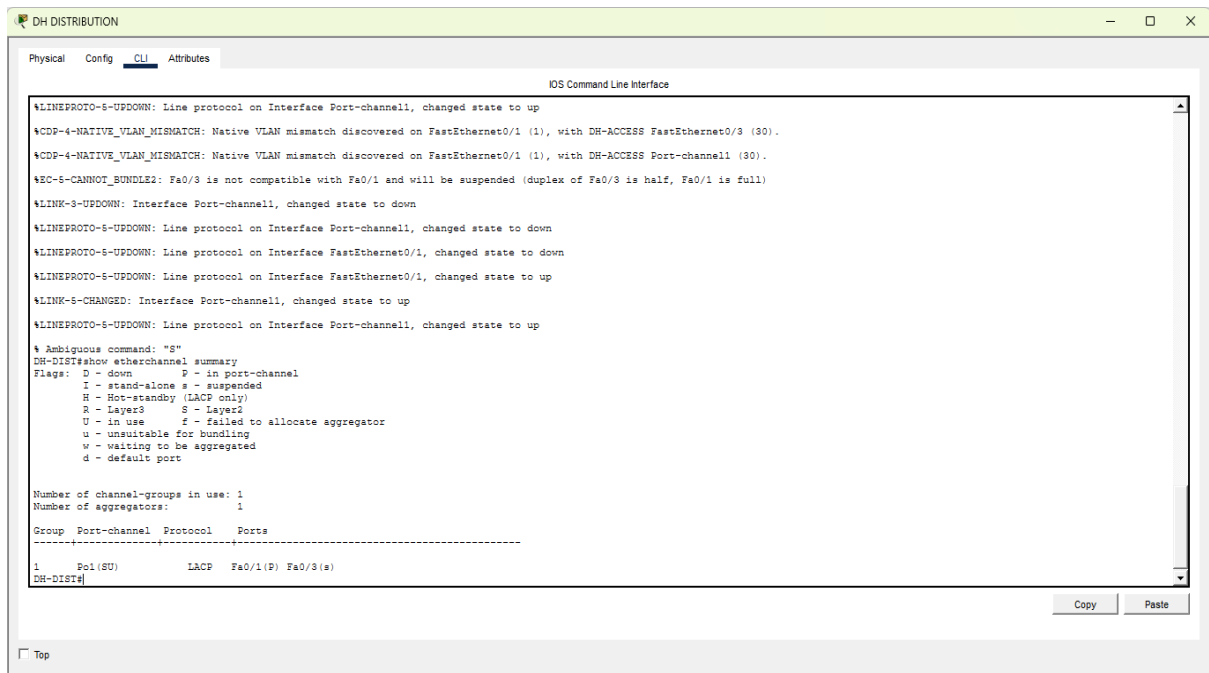


The screenshot shows a terminal window titled "DH DISTRIBUTION" with tabs for Physical, Config, CLI, and Attributes. The CLI tab is active, displaying the "IOS Command Line Interface". The terminal output shows the following commands and responses:

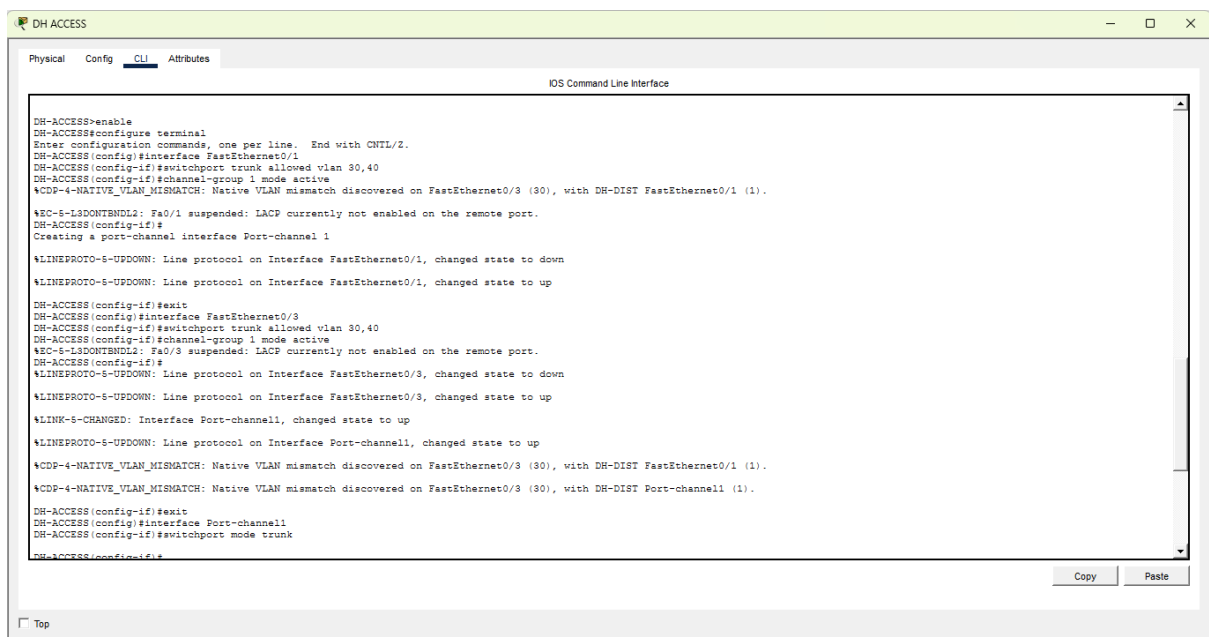
```
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with DH-ACCESS FastEthernet0/3 (30).
% Incomplete command.
DH-DIST(config)#interface FastEthernet0/3
DH-DIST(config-if)#switchport trunk encapsulation dot1q
DH-DIST(config-if)#switchport trunk allowed vlan 30,40
DH-DIST(config-if)#channel-
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with DH-ACCESS FastEthernet0/3 (30).
% Ambiguous command: "channel-"
DH-DIST(config-if)#channel-group 1 mode active
%EC-5-L3CONTBNDL2: Fa0/3 suspended: LACP currently not enabled on the remote port.
DH-DIST(config-if)#
%EC-5-CANNOT_BUNDLE2: Fa0/3 is not compatible with Fa0/1 and will be suspended (duplex of Fa0/3 is half, Fa0/1 is full)
DH-DIST(config-if)#exit
DH-DIST(config)#interface Port-channel1
DH-DIST(config-if)#switchport trunk encapsulation dot1q
% Invalid input detected at '^' marker.
DH-DIST(config-if)#switchport trunk encap
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with DH-ACCESS FastEthernet0/3 (30).
% Incomplete command.
DH-DIST(config-if)#switchport trunk allowed 30,40
% Invalid input detected at '^' marker.
DH-DIST(config-if)#switchport trunk allowed vlan 30,40
DH-DIST(config-if)#end
DH-DIST#
%SYS-5-CONFIG_I: Configured from console by console
DH-DIST#write memory
Building configuration...
[OK]
DH-DIST#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with DH-ACCESS FastEthernet0/3 (30).
```

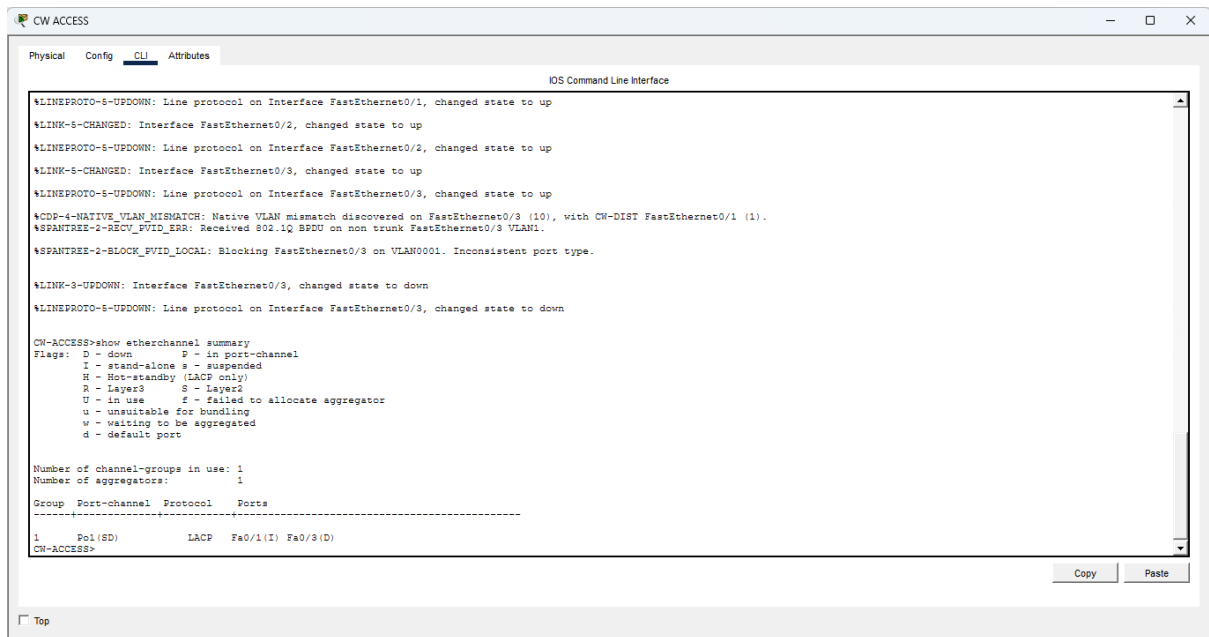
At the bottom right of the terminal window, there are "Copy" and "Paste" buttons. A "Top" link is located at the bottom left of the window.

Result of EtherChannel Configuration:

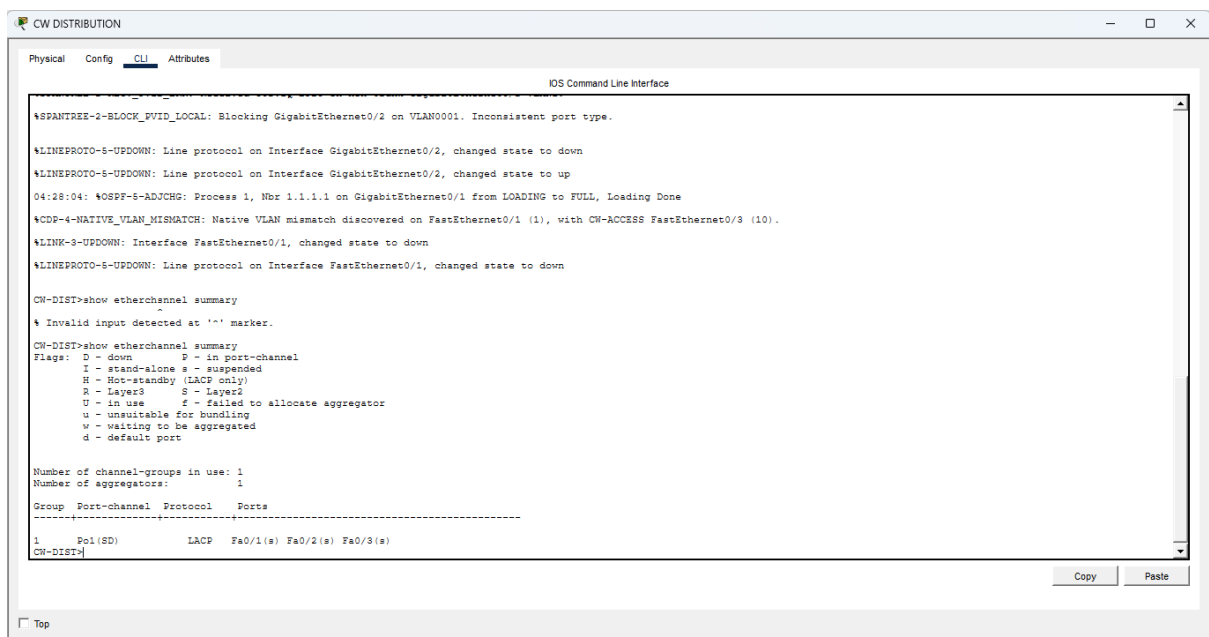


DH ACCESS:





CW DISTRIBUTION:



During the failure the connection between CW DISTRIBUTION and CW ACCESS was tested with FastEthernet links rather than GigabitEthernet links since those were the only options we had. Due to only having FastEthernet links we had to use 'switchport trunk encapsulation dot1q' to fix the VLAN mismatches we kept facing. The port security ports shifted to avoid fa0/4 – 15 to overlap with the etherchannel ports.

The network recovered through the fa0/3 etherchannel link. Traffic continued to flow without manual configuration.